Diffusion Proposals for Redistricting Markov Chains

Alaittin Kırtışoğlu
Department of Applied Mathematics, Illinois Institute of Technology
Chicago, IL 60616
kirtisoglu@hawk.iit.edu

October 22, 2026

Abstract

Recombination (ReCom) chains are the standard Markov chain Monte Carlo tool for sampling redistricting plans, and they are routinely run under a Metropolis filter that nominally targets a compactness-tempered distribution. We introduce a proposal built from graph diffusion. One convex energy on the merged pair of districts, a kernel-weighted Laplacian smoothness term plus a fidelity term that anchors each move to the current plan, is minimized under global population balance and rounded at random. The diffusion view yields a small theory of the proposal: its optimum is a single linear solve whenever balance is slack, the expected number of units a proposal moves is an explicit high-pass filter norm with a provable ceiling, and mobility is governed by feasibility rather than by the Metropolis ratio, which gives a tuning rule. Correcting the rounding with an exact Metropolis–Hastings step gives, to our knowledge, the first optimization-based redistricting proposal with a detailed-balance proof. Mixed with the forest-lifted recombination move Metropolized to the same target, it yields a reversible sampler for kernel-weighted compactness distributions, validated against enumerated targets. On Iowa it roughly doubles the effective samples per second of the forest move alone under perimeter and density targets, and adds nothing on the spanning-tree-weighted measure. The reason is structural, and we measure it: the tree move is an independence sampler from the untempered spanning-tree measure, so its acceptance decays exponentially with the size of the merged pair under any compactness-tempered target, while the local move has a dial for the same effect. On an enumerated test bed we also show, at the level of distributions, that plain and tempered ReCom do not sample their nominal target and sit close to the spanning-tree-weighted measure.

1 Introduction

1.1 The redistricting problem and ReCom's role

Political redistricting seeks a partition of a geographic region into electoral districts satisfying legal, demographic, and geometric requirements, namely contiguity, population balance, compactness, and preservation of communities of interest. Computational approaches model the problem on the dual adjacency graph $G = (V, E)$ of geographic units (precincts, census blocks, counties). A valid k-district plan is a partition

$$P = \{V_1, \dots, V_k\}, \qquad \bigsqcup_{i=1}^{k} V_i = V,$$

where each $G[V_i]$ is connected and the populations $p(V_i)$ are approximately equal. Let $\mathcal{P}_k(G)$ denote the space of valid k-district plans. The cardinality of $\mathcal{P}_k(G)$ is enormous. Even four congressional districts over the 99 counties of Iowa admit roughly 10^{24} plans [28].

Markov chain Monte Carlo (MCMC) has become the standard tool for sampling $\mathcal{P}_k(G)$. Early flip chains modify a single precinct's assignment per step [27] and mix slowly. The Recombination (ReCom) algorithm of DeFord, Duchin, and Solomon [22] performs multi-precinct updates by merging two adjacent districts, sampling a uniform random spanning tree of the merged subgraph, and cutting an edge to recover a balanced bipartition. ReCom dramatically improved the mixing efficiency of redistricting MCMC and is now the workhorse of the field, with reversible [11], forest-lifted [4] and multi-scale [3] extensions.

ReCom's blind spot. The uniform spanning-tree split treats every edge of the merged subgraph identically. The proposed bipartition is determined by the topology of the merged subgraph alone. Real geography is not topology-only. Shared boundary length, population density, demographic similarity, and administrative boundaries all carry signal that could shape the proposed split. Practitioners have felt this. County-weighted trees [19] and GerryChain's region-aware surcharges [40] inject indicator-based edge weights into the tree draw, and SpecReCom [21] replaces the tree with a Fiedler-vector cut of a randomly weighted Laplacian. But these mechanisms are restricted (region indicators, hard-coded county kernels, or no kernel at all), none includes a locality-preserving fidelity term, and, as their authors acknowledge, each shifts the chain's stationary distribution in an uncharacterized, uncorrected way.

A second blind spot, which this paper measures. Even the exact versions of the tree move, which Metropolize it to a stated target, propose from one fixed law, the spanning-tree measure on balanced splits of the merged pair. Whenever the target a practitioner wants is not close to that law, for instance a compactness-tempered target or a geography-weighted one, the Metropolis step has to undo the mismatch on every proposal, and the mismatch grows with the size of the merged pair. We show that the resulting acceptance decays exponentially in the pair size, which is the structural reason an exact sampler for such targets needs a second kind of move.

1.2 Contribution

We introduce a diffusion proposal for ReCom-style MCMC and the exact sampler built on it. The core object is a single Laplacian energy on the merged-pair subgraph $H = G[V_i \cup V_j]$,

$$E(x) \;=\; \alpha\, x^\top L\, x \;+\; \beta\, \|x - x_0\|^2, \qquad (1-\varepsilon)\bar{p} \;\le\; p^\top x \;\le\; (1+\varepsilon)\bar{p}, \tag{1}$$

where L is the kernel-weighted graph Laplacian of H, x_0 encodes the current bipartition, and the balance constraint is anchored to the global per-district ideal (Equation (4)). The Laplacian term $x^\top Lx$ counts (or approximates) cut edges between the two sides; the fidelity term $\|x - x_0\|^2$ penalizes deviation from the current assignment, enforcing locality of each move. The paper makes four contributions.

1. **A diffusion proposal with a theory of its own behavior** (§4.2–§4.3). The interior optimum of the energy is one implicit Euler step of the graph heat equation with timestep $\tau = \alpha/\beta$. We show that this step is a single linear solve whose output lies in the unit cube automatically, so the box constraints never bind and the proposal costs a linear solve whenever population balance is slack (Lemma 4.1). Under single-shot Bernoulli rounding, the expected number of units a proposal moves is exactly the one-norm of a high-pass filter applied to the current split (Proposition 4.2). That quantity grows like twice the timestep times the pair's boundary for small timesteps and is bounded above by half the merged pair for any timestep (Corollary 4.1). Given a feasible proposal the Metropolis step accepts it at least 98 percent of the time; the chain's stickiness is a feasibility effect with an interior optimum in τ (§4.6), and its mobility is given to within counting noise by a two-swap law, Equation (10).

2. **An exactly reversible correction** (§4.5). Correcting the rounding with an exact Metropolis–Hastings step yields Chain E, to our knowledge the first optimization-based redistricting proposal

with a detailed-balance proof (Proposition 4.3), stated on labeled plans, where the proposal map is a bijection. We verify the implemented kernel to machine precision on an enumerated fiber and show by a batch-means test that a million-step run's occupancy distribution is not distinguishable from the exact target (§5.2).

3. **An exact sampler for kernel-weighted compactness targets, and a measurement of when its local component matters** (§4.7, §5.3–§5.5). We lift Chain E to the spanning-forest state space and mix it with the forest-lifted recombination move of Autry et al. [4], Metropolized to an arbitrary kernel-weighted target. The mixture is verified to machine precision on an enumerated lifted fiber and against enumerated targets on 4 006 plans. On Iowa, under a criterion fixed before the runs, the mixture gives 1.6 to 2.3 times the forest move's effective samples per second under perimeter and density targets on two seeds, and 0.8 to 0.9 times under the spanning-tree-weighted tempered measure. We then show why. The forest move is an independence sampler from the untempered spanning-tree measure on balanced splits; on $N \times N$ grids its acceptance at equilibrium falls like $e^{-0.24N}$ under compactness-tempered targets, with or without tree weighting, while the per-proposal mismatch in its Hastings ratio grows in proportion to the boundary length. The local move loses acceptance with pair size too, but has a dial for it, the move-size law; the tree move has none.

4. **A ground-truth calibration of what the ReCom family samples, at the level of distributions** (§5.1, §5.6). On a fully enumerated 5×5 fiber we compute the spanning-tree-weighted measure for every state by the Matrix–Tree theorem and compare full empirical distributions against it and against the nominal tempered target, with sampling-noise floors at the effective sample size. The tempered chain's cut distribution is 220 floor standard deviations from its nominal target and close to, though distinguishable from, the tree-weighted tempered measure, while the exact chain is indistinguishable from the nominal target. On Iowa the exact sampler puts the nominal target's mean cut count near 88 and the tree-weighted measure's at 41, where tempered ReCom sits at 40.

Positioning. The construction of Chain E, a deterministic optimization that sets the parameters of a random proposal which is then corrected exactly, has ancestors outside redistricting. Randomize-then-optimize [5] samples posteriors by perturbing and solving, and Zanella's informed proposals [56] give the general theory of locally balanced, target-aware proposals on discrete spaces. Chain E is an instance of that program on the space of plans, and our priority claim is scoped to redistricting accordingly. Within redistricting, the known-target samplers, Metropolized forest recombination [4], RevReCom, Cycle Walk, Marked Edge Walk, and BUD [11, 24, 38, 1], achieve computable transition probabilities by lifting to spanning-forest state spaces, while Chain E achieves the same property at the partition level; the two routes are complementary, and the forest move is the global component of our sampler. Bonsai sampling [20] draws independent plans from a specified law without a chain and is the natural non-MCMC comparison for any known-target method. §2 positions the proposal against the weighted-tree, spectral, and matheuristic lines.

Roadmap. §2 reviews related work and §3 sets up notation. §4 develops the proposal, its theory, the exact chain, and the sampler; §5 reports the calibration, the convergence validation, the kill test against the forest move, the scaling experiment, and the Iowa calibration. Appendices hold hyperparameters, a partition-fiber diagnostic, and a reproducibility checklist.

2 Related work

MCMC for redistricting. Early MCMC for redistricting framed the problem as a graph cut and used single-unit flip chains [27] or Swendsen–Wang-style large-swap moves. Flip chains mix poorly and produce highly irregular districts [22, 41]. ReCom [22] introduced the spanning-tree merge-split move and is now standard. Extensions include Multi-Scale Merge–Split [3] preserving nested communities (counties, precincts), Metropolized forest recombination [4], Sequential Monte Carlo [36] and its generalized form [42], whose repetition effects are analyzed in [10], multiscale parallel tempering [16], and reversible recombination [11] targeting the canonical spanning-tree distribution. The compactness-tempered target that practitioners run under a Metropolis filter, and that we audit here, originates in the North Carolina analyses of Herschlag et al. [32]. The 50-State Redistricting Atlas [37] is the field's standard ensemble benchmark. A consistent theme of the 2022–2026 literature is that a sampler is only as defensible as its target. RevReCom [11], Metropolized forest recombination [4], the cycle walk [24], and the marked edge walk [38] all pay a structural price (lifted state spaces, spanning-forest counting) precisely to keep transition probabilities computable. On the theory side, Cannon, Pegden, and Tucker-Foltz [13] give a polynomial-time sampler for the spanning-tree measure on balanced partitions of grids with a constant number of districts, Gold [30] exhibits planar-grid configurations from which recombination needs exponentially many expected steps to re-split, and mixing-time bounds for the local chains used in practice remain open. Akitaya et al. [2] prove that deciding reachability between balanced partitions under recombination is PSPACE-complete when the district count and step size grow with the graph.

Weighted and geography-aware proposals. Edge weighting inside the ReCom proposal is not new, and we position our work against it explicitly. Clelland et al. [19] replace the proposal's edge-weighting function with *county-weighted trees* (intra-county edge weights drawn from the uniform distribution $\mathcal{U}[0, w]$ for a tunable scalar w) to reduce county splits in Colorado ensembles. GerryChain's region-aware ReCom [40] implements the same idea. User-specified surcharges are added to the randomized Kruskal weights for edges crossing designated region boundaries. Tapp [49] modifies the ReCom tree-and-cut step to improve shape-based compactness at the proposal level. In all three, the weighting is an indicator- or attribute-indexed heuristic, and (as their authors acknowledge) the induced stationary distribution is uncharacterized and uncorrected. What the unweighted tree proposal itself induces is better understood. Clelland et al. [18] measured ReCom ensembles on grids and precinct graphs and found the sampling probability of a plan to decay roughly exponentially in its cut length, and Procaccia and Tucker-Foltz [44] proved the underlying fact that compact plans have many spanning trees. Tapp, Proebsting, and Ramsay [50] study how tolerance and county-preservation settings move ReCom ensembles and propose effective-sample-size diagnostics for them. Our calibration in §5.1 makes the first of these observations exact on an enumerated fiber, at the level of full distributions, and our scaling experiment in §5.5 turns the second into a statement about acceptance: the same exponential that makes compact plans tree-rich is what an exact tree move has to pay, per proposal, to target anything else. Conversely, the samplers with known targets cited above handle only unweighted spanning-tree/forest measures or other tree-count-based families. The gap this paper addresses sits at the intersection, namely a proposal that accepts an *arbitrary* nonnegative edge kernel as a first-class input together with an exact accept/reject step against a stated target (§4.5, §4.7).

Optimization-based districting and matheuristics. A parallel operations research (OR) literature treats districting as exact optimization. Examples include the Hess model lineage, column-generation and set-partitioning formulations [31], multilevel multiobjective methods [48], contiguity-constrained MIPs [53], and cut-edge minimization solved to optimality at county/tract scale [52]. Dobbs et al. [25] solve for minimal-change and midpoint plans under transfer distance to reference plans exactly by mixed-

integer programming. Between the sampling and optimization families [6], hybrids are emerging. Short bursts [12] use ReCom as a stochastic optimizer; Brous and Shmoys [8] replace the spanning-tree re-split with a Gurobi integer program inside an accept-if-better local search for majority-minority districts. The integer form of our energy, a convex MIQP refinement inside a recombination chain, belongs to this matheuristic line; we note in §6.5 what we found when we evaluated it as an optimizer, and do not pursue it here. The proximal term $\beta\|x - x_0\|^2$ is the quadratic analogue of local branching [29], imposing a soft trust region around the incumbent.

Spectral partitioning and diffuse-interface models. Spectral graph partitioning has a long history in image segmentation and clustering [45]. In redistricting, the recent SpecReCom of Davies, Job, Kampbell, Kim, and Seo [21] (arXiv preprint, June 2025) replaces the random spanning tree with a Fiedler-vector sign cut on a randomly edge-weighted Laplacian of the merged pair, reporting roughly half the cut edges of RevReCom on a 56×56 grid and a Colorado planar graph. Plain SpecReCom sacrifices population balance, which their BalSpecReCom variant partially restores. Diffuse-interface models build on the Ginzburg–Landau (GL) functional, combining a smoothness term with a fidelity term; the Chan–Vese model [14] and Merriman–Bence–Osher (MBO) threshold dynamics implement this iteratively. Bertozzi and collaborators [7, 39] port the machinery to weighted graphs, where the smoothness term becomes the Laplacian quadratic form $x^\top Lx$ and graph Tikhonov regularization [43, 46] recovers heat-kernel smoothing. Jacobs and Walch [34] brought threshold dynamics to redistricting before SpecReCom, using volume-constrained MBO via auction dynamics [33] to generate compact equal-population maps. Theirs is a continuum PDE approach, not a graph-kernel one, and it is not embedded in MCMC. To our knowledge the present work is the first to use the *graph* GL lineage (kernel-weighted Laplacian smoothing plus fidelity) as a proposal inside redistricting MCMC.

Cut minimization and discrete compactness. The minimum graph cut [47, 35] and its balanced variants such as normalized cut [45] are classical objectives. NP-hardness of balanced minimum cut [54] motivates continuous relaxations, of which the Fiedler-vector cut and the Total Variation Isoperimetric Profile [23] are two well-known examples. The cut edge count $|\partial P|$ has been proposed as a discrete compactness measure for redistricting [26] and shown to correlate well with classical geometric scores (Polsby–Popper, Reock) while being more stable under unit refinement.

3 Background: plans and ReCom

3.1 Dual graph and k-district plans

Let $G = (V, E)$ be the dual graph of a state, with vertices representing geographic units (precincts, census blocks, counties) and edges encoding shared boundaries. Each vertex v carries a population $p_v \geq 0$ and may carry additional attributes (demographic vectors, shapefile geometry). A *k-district plan* is a partition $P = \{V_1, \ldots, V_k\}$ of V such that each induced subgraph $G[V_i]$ is connected (*contiguity*) and the populations satisfy

$$(1 - \varepsilon)\bar{p} \;\leq\; p(V_i) \;\leq\; (1 + \varepsilon)\bar{p}, \qquad \bar{p} := \frac{p(V)}{k}, \tag{2}$$

for a tolerance $\varepsilon \in [0, 1)$. We write $\mathcal{P}_k(G; \varepsilon)$ for the set of valid plans. The *cut edges* of a plan are

$$\partial P \;=\; \{(u, v) \in E : P(u) \neq P(v)\},$$

and we call $|\partial P|$ the *cut-edge count* of P. This quantity is the discrete analog of the perimeter of the district boundaries and serves as our primary compactness metric [26]. For a nonnegative edge kernel w we write $w(\partial P) = \sum_{e \in \partial P} w_e$ for the *weighted cut*.

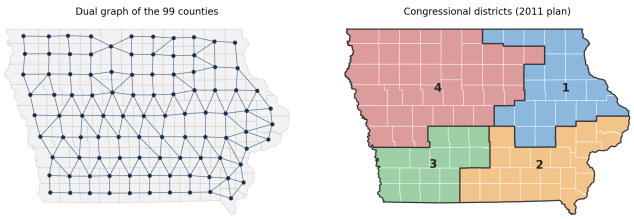


Figure 1: The dual graph of Iowa's 99 counties (left) and the four congressional districts enacted after the 2010 census (right). Nodes sit at county centroids; two counties are joined by an edge when they share a boundary segment.

3.2 Recombination (ReCom)

Recombination [22] is a Markov chain on $\mathcal{P}_k(G;\varepsilon)$ whose single step is given in Algorithm 1.

Algorithm 1: Vanilla ReCom step (Chain V)

Input: Current plan P, dual graph G, balance tolerance ε.

Output: Next plan P'.

Select two adjacent districts V_i, V_j uniformly at random from boundary pairs;

Form the induced subgraph $H = G[V_i \cup V_j]$;

repeat

  Sample a uniform random spanning tree T of H via Wilson's algorithm [55];

  Search edges of T for a cut e^* whose removal produces two ε-balanced components.

until *a cut e^* exists yielding two ε-balanced components*;

Define V_i', V_j' as the two components of $T \setminus \{e^*\}$;

$P' \leftarrow P$ with V_i, V_j replaced by V_i', V_j';

return P'.

The tree-based proposal flips many precincts simultaneously between V_i and V_j while preserving contiguity automatically (any edge removal from a tree produces two connected components). This is the key advantage over single-precinct flip chains, which traverse the state space too slowly to be practically useful.

Chains compared. Table 1 lists the chains of this paper.

Chain	Method	Mechanism	Stationarity claim
V	Vanilla ReCom [22]	spanning-tree split, always accept	unknown target
A	Tempered ReCom	ReCom + MH on cuts (symmetric-q)	approximate
E	QP-MH (this work, §4.5)	QP + Bernoulli rounding + exact MH	**exact**
T	Partition-level tree move (§4.7.1)	uniform tree, uniform edge, tree-count MH	**exact**
F	Forest move (§4.7.2)	forest-lifted ReCom, balanced edge, exact MH	**exact**
H	Hybrid (this work, §4.7)	mixture of F and lifted E, any kernel-weighted target	**exact**

Table 1: The chains compared in this paper. The last column is the chain's relationship to its stated target. Chains E, T, F and H carry exact Metropolis–Hastings corrections; Chain A relies on the symmetric-q approximation (§4.5) and Chain V has no stated target. Chain T is kept as the reference that explains Chain A and as the baseline of an earlier version of this work; Chain F is the baseline the sampler is judged against.

Tempered ReCom uses a Metropolis–Hastings filter targeting $\pi(P) \propto \exp(-\lambda|\partial P|)$ with the symmetric-

q approximation (treating the forward and reverse proposal probabilities as equal); see Appendix A for the locked-in $\lambda = 0.10$. All chains enforce the global two-sided balance window of Equation (4) on every move; we note (§5) that the raw GerryChain `recom` proposal alone does not.

The proposal we replace. The tree-and-cut step is kernel-blind (§1.1), and the cut edge is chosen only after the tree is drawn, so the proposal cannot anticipate the shape of the split. The diffusion proposal of §4 replaces that step with a single convex optimization of a kernel-weighted Laplacian energy plus a locality-preserving fidelity term. The energy $x^\top L\, x = \sum_{(u,v)\in E} w_{uv}(x_u - x_v)^2$ equals the weighted cut count for binary x, so minimizing it directly minimizes the discrete perimeter in the chosen kernel.

4 The QP-diffusion model, its theory, and the sampler

The energy (1) has three ingredients: the kernel that defines the Laplacian (§4.1), the convex program on the merged pair (§4.2, with its spectral form (7)), and the balance window. The rest of the section builds the chains on top of it.

4.1 Locality kernels

Choosing the edge weights w_{ij} is the most important modeling decision. The kernel determines what *smoothness* means and therefore controls how district boundaries move under the relaxation. We restrict all weights to geographic neighbors,

$$w_{ij} = 0 \quad \text{if } (v_i, v_j) \notin E, \tag{3}$$

ensuring that diffusion operates only through geographic adjacency and preventing long-range interactions that could produce disconnected districts.

Uniform kernel (k_{unif}). $w_{ij} = 1$ for every edge. The Laplacian L is the combinatorial graph Laplacian; $x^\top Lx$ is exactly the cut edge count. In the $\beta \to 0^+$ limit the proposal approaches a balanced Fiedler-style split.

Perimeter kernel (k_{perim}). $w_{ij} = |\partial v_i \cap \partial v_j|$ (shared-boundary length). The Laplacian quadratic form $x^\top Lx$ is then the *geographic perimeter* of the partition, weighted by boundary segment length. This kernel discourages fragmented districts and favors boundaries that cut along short segments.

Density kernel (k_{dens}). Let ρ_v denote the population density at vertex v (population per unit area). A Gaussian density kernel

$$w_{ij} \;=\; |\partial v_i \cap \partial v_j| \cdot \exp\Big(-\frac{(\rho_i - \rho_j)^2}{\sigma^2}\Big)$$

encourages cuts along sharp density gradients (urban–rural boundaries) while preserving the topological role of shared boundary length.

Demographic kernel (k_{demog}). Let $d_v \in \mathbb{R}^m$ be a vector of demographic attributes. A Gaussian similarity kernel $w_{ij} = \exp(-\|d_i - d_j\|^2/\sigma^2)$ encourages communities of interest. We define this kernel for completeness but do not use it in our experiments, and a caution stronger than "use as a soft regularizer" is warranted. If the attribute vector encodes race, a proposal that systematically shapes district boundaries around racial similarity risks making race the predominant districting criterion, which triggers strict scrutiny under the *Shaw v. Reno* line of racial-gerrymandering cases. Any practical use of demographic kernels should be reviewed against that doctrine, should document the kernel's weight relative

to race-neutral criteria, and is more defensible for attributes that are legitimate districting factors in the jurisdiction (e.g., municipal membership, communities of interest as defined by state law).

Composite kernels. Linear combinations $w_{ij} = \sum_k a_k w_{ij}^{(k)}$ with $a_k \geq 0$ remain valid, since they preserve nonnegativity and locality.

4.2 The convex QP

Let $H = G[V_i \cup V_j]$ be the merged subgraph of two adjacent districts. Let L_H be the kernel-weighted Laplacian of H, let $x_0 \in \{0,1\}^{|V_H|}$ encode the current bipartition (1 for vertices in V_i, 0 for vertices in V_j), and let p be the population vector restricted to H. Fix weights $\alpha, \beta > 0$ and balance tolerance $\varepsilon \in (0,1)$. Crucially, the balance constraint is anchored to the *global* per-district ideal $\bar{p} = p(V)/k$ from Equation (2), not to $p(V_H)/2$. Both sides of the split must individually satisfy the plan-validity bound, which yields the two-sided window

$$\ell \;=\; \max\big((1-\varepsilon)\bar{p},\; p(V_H) - (1+\varepsilon)\bar{p}\big), \qquad u \;=\; \min\big((1+\varepsilon)\bar{p},\; p(V_H) - (1-\varepsilon)\bar{p}\big). \tag{4}$$

Anchoring to $p(V_H)/2$ instead (a tempting simplification, and one present in an earlier version of our own implementation) only bounds the local split. Because $p(V_H)$ itself drifts as the chain moves population between pairs, district populations can ratchet outside the global tolerance by a factor of up to $(1+\varepsilon)$ per move on heterogeneous graphs. Our experiments audit the global bound at every step (§5).

Definition 4.1 (Continuous QP-diffusion)**.** *The continuous QP-diffusion bipartition of H is*

$$x^\star \;=\; \arg\min_{x \in [0,1]^{|V_H|}} \; \alpha\, x^\top L_H\, x \;+\; \beta\, \|x - x_0\|_2^2 \tag{5}$$

$$\text{subject to } \; \ell \;\leq\; p^\top x \;\leq\; u, \tag{6}$$

with ℓ, u as in Equation (4).

The first term in (5) is the kernel-weighted Laplacian smoothness energy. For binary x it equals the weighted cut count $\sum_{(u,v)\in E(H)} w_{uv}\, \mathbf{1}[x_u \neq x_v]$, and the continuous relaxation $x \in [0,1]^{|V_H|}$ extends this to a smooth objective on the unit cube. The second term is the *fidelity term*, which penalizes deviation from the current bipartition and ensures that each proposed move is local. The balance constraint enforces ε-balance on the proposed split.

Proposition 4.1 (Convexity and well-posedness)**.** *The objective in (5) is strictly convex on $[0,1]^{|V_H|}$ (since $\beta > 0$ and L_H is positive semidefinite). The feasible region defined by (6) is a nonempty polytope whenever the current plan is valid (it then contains x_0). Hence the QP admits a unique minimizer $x^\star$, computable in polynomial time by any convex QP solver.*

Spectral interpretation. Projecting the interior optimality condition $(\alpha L_H + \beta I)x^\star = \beta x_0$ onto the eigenbasis $\{\phi_k\}$ of L_H with eigenvalues $0 = \lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_n$ yields the closed-form frequency response

$$\hat{x}_k^\star \;=\; \frac{\beta}{\alpha\lambda_k + \beta}\, \hat{x}_{0,k}, \qquad \hat{y}_k := \phi_k^\top y. \tag{7}$$

The QP therefore acts as a low-pass spectral filter on the prior x_0, with cutoff controlled by the ratio α/β. Equivalently, with $\tau := \alpha/\beta$, the optimum satisfies $(x^\star - x_0)/\tau = -L_H x^\star$, the implicit Euler discretization of the graph heat equation $\dot{x} = -L_H x$. Inspecting the leading eigenvectors $\phi_2, \phi_3, \ldots$ provides a sanity check on kernel choice. If they look like noise rather than coherent geographic regions, the chosen kernel is probably not informative.

4.3 Theory of the diffusion proposal

The spectral identity is usually read as interpretation. Worked through, it predicts how the proposal behaves, and every prediction below is checked against the solver. Write $\tau := \alpha/\beta$ for the diffusion timestep, $L = L_H$ for the kernel-weighted Laplacian of the merged pair, $n = |V_H|$, and $\mathbf{1}_i \in \{0,1\}^n$ for the indicator of V_i, so that $x_0 = \mathbf{1}_i$.

Lemma 4.1 (The box constraints never bind)**.** *For any nonnegative kernel and any $\tau > 0$, the matrix $(I + \tau L)^{-1}$ is symmetric, entrywise nonnegative, and has unit row sums. Hence the unconstrained minimizer $x^\star = (I + \tau L)^{-1} x_0$ of (5) is a convex combination of the entries of x_0 and lies in $[0,1]^n$, and the QP of Definition 4.1 reduces to this single linear solve whenever the balance constraint (6) is slack at $x^\star$.*

Proof. $I + \tau L$ is symmetric with positive diagonal, nonpositive off-diagonal entries, and row sums equal to one, so it is strictly diagonally dominant, hence a nonsingular M-matrix, and its inverse is entrywise nonnegative. Since $L\mathbf{1} = 0$ we have $(I + \tau L)\mathbf{1} = \mathbf{1}$ and so $(I + \tau L)^{-1}\mathbf{1} = \mathbf{1}$. Each row of the inverse is therefore a probability vector and $x^\star_v$ is a weighted average of entries of $x_0 \in \{0,1\}^n$. The objective is strictly convex, so if the unconstrained minimizer satisfies the one remaining constraint it is the constrained minimizer. □

Proposition 4.2 (Expected move size)**.** *Under the single-shot Bernoulli rounding of §4.5, and whenever the balance constraint is slack, the expected number of units of H that a proposal reassigns is*

$$\mathbb{E}[\#\text{flips}] \;=\; \|x^\star - x_0\|_1 \;=\; 2\,\mathbf{1}_i^\top (I + \tau L)^{-1} \mathbf{1}_j \;=\; \frac{2\,|V_i|\,|V_j|}{n} \;-\; 2 \sum_{k \geq 2} \frac{(\phi_k^\top \mathbf{1}_i)^2}{1 + \tau \lambda_k}, \tag{8}$$

where (λ_k, ϕ_k) are the eigenpairs of L with $\lambda_1 = 0$.

Proof. A unit with $x_{0,v} = 0$ flips with probability $x^\star_v$ and one with $x_{0,v} = 1$ with probability $1 - x^\star_v$; both equal $|x^\star_v - x_{0,v}|$ because $x^\star_v \in [0,1]$ by Lemma 4.1, which gives the first equality. Write $P = (I + \tau L)^{-1}$. For $v \in V_i$, $1 - x^\star_v = \sum_{u \in V_j} P_{vu}$ by unit row sums, and for $v \in V_j$, $x^\star_v = \sum_{u \in V_i} P_{vu}$; summing and using symmetry of P gives $2\,\mathbf{1}_i^\top P\,\mathbf{1}_j$. Expanding $P = \sum_k \phi_k \phi_k^\top / (1 + \tau\lambda_k)$ with $\phi_1 = \mathbf{1}/\sqrt{n}$ gives the $k = 1$ term $|V_i||V_j|/n$, and since $\phi_k \perp \mathbf{1}$ for $k \geq 2$, $\phi_k^\top \mathbf{1}_j = -\phi_k^\top \mathbf{1}_i$, which gives the sum. Clipping at $\delta = 10^{-12}$ changes the expectation by at most δn. □

Corollary 4.1 (Small-timestep law, monotonicity, and ceiling)**.** *Let $|\partial_w(V_i, V_j)|$ be the boundary of the pair measured in the proposal's kernel. Then $\mathbb{E}[\#\text{flips}]$ is increasing in τ,*

$$\mathbb{E}[\#\text{flips}] \;=\; 2\tau\,|\partial_w(V_i, V_j)| + O(\tau^2) \quad (\tau \to 0), \qquad \mathbb{E}[\#\text{flips}] \;\leq\; \frac{2\,|V_i|\,|V_j|}{n} \quad \textit{for every } \tau,$$

with equality in the limit $\tau \to \infty$. For a balanced pair the ceiling is $n/2$, half the merged pair.

Proof. Each subtracted term in (8) is nonnegative and decreasing in τ, which gives monotonicity and the bound, and the terms vanish as $\tau \to \infty$. For the small-τ law, expand $(1 + \tau\lambda_k)^{-1} = 1 - \tau\lambda_k + O(\tau^2)$; the τ^0 terms cancel because $\sum_{k \geq 2} (\phi_k^\top \mathbf{1}_i)^2 = |V_i| - |V_i|^2/n = |V_i||V_j|/n$, and the τ^1 term is $2\tau \sum_k \lambda_k (\phi_k^\top \mathbf{1}_i)^2 = 2\tau\,\mathbf{1}_i^\top L\,\mathbf{1}_i = 2\tau |\partial_w|$. □

Three consequences follow. The continuous proposal is far cheaper than "solve a QP" suggests, since it is one linear solve in the common case. The move size is a design dial with a known scale, so to propose moves of about m units one sets $\alpha/\beta \approx m/(2|\partial_w|)$ while the small-timestep law holds; §5.5 uses this dial. And the dial has a hard stop. No choice of α and β can lift the expected move above half the merged pair, so mobility cannot be bought by tuning, which is the structural reason a global move is needed alongside

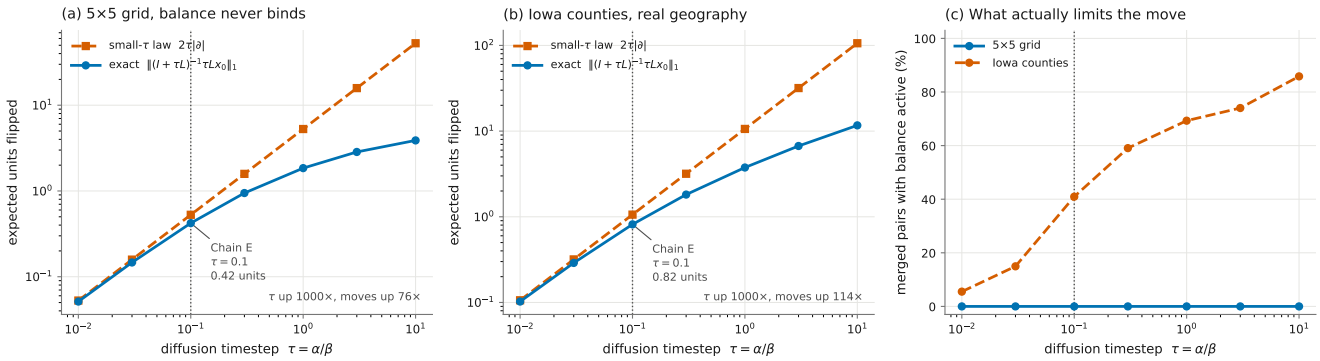


Figure 2: The move-size law against the solver. (a) On the enumerated 5×5 fiber, 1 411 merged pairs drawn from 200 plans; the exact filter norm of (8) and the small-timestep law $2\tau|\partial_w|$ agree for $\tau \leq 0.03$ and separate as the gain saturates. Raising τ by $1000\times$ raises the move by $76\times$, and the last decade buys $2.1\times$. The locked Chain E setting $\tau = 0.1$ is marked. (b) The same on Iowa, 127 pairs from 25 tree-partition plans. (c) Fraction of merged pairs whose balance constraint is active. On the uniform grid it never is, so the proposal is a linear solve at every step; on real county geography it is active for 41% of pairs at $\tau = 0.1$ and 86% at $\tau = 10$, so there the population constraint, not the diffusion, caps the move.

the exact local chain (§4.6, §4.7). The kernel enters through $|\partial_w|$, so a perimeter kernel takes larger steps exactly where the geographic border is long.

Verification. Across 10 766 linear solves on the two test beds the unconstrained optimum never left the unit cube, as Lemma 4.1 requires. On 120 merged pairs with balance slack, Gurobi's constrained solution and the linear solve agree to 1.1×10^{-4}, the solver tolerance, so the implementation matches the theory. The median relative error of the small-timestep law against the exact filter norm is 2.5% at $\tau = 0.01$, 7.4% at 0.03, and 24% at 0.1 (Figure 2). At the paper's locked Chain E setting the expected move is 0.42 units on the 5×5 fiber and, on Iowa, 0.82 on the pairs whose window stays slack at every timestep and 1.26 averaged over all 127 pairs, about one unit per proposal, which is the quantitative content of the chain's stickiness reported in §5.1.

4.4 Deterministic rounding, and why it blocks the correction

The continuous optimum $x^\star \in [0,1]^{|V_H|}$ must be rounded to a bipartition. The natural deterministic choice is a Cheeger sweep [17], which sorts the entries of $x^\star$, scans the thresholds, keeps the balanced and connected candidate with the smallest cut, and repairs near-misses by flipping orphaned components to the other side. As a proposal it has a defect no repair fixes. An argmin-plus-sweep map is deterministic, so the reverse proposal has zero probability and the Hastings ratio degenerates; a chain built on it is a heuristic with no stationarity claim beyond ergodicity. §4.5 replaces the sweep with single-shot randomized rounding, which restores an everywhere-positive proposal density and an exact correction. The same defect, in a different guise, is why the forest move of §4.7.2 must keep the tree draw and the cut-edge draw random rather than optimized.

4.5 The exactly reversible QP-MH chain (Chain E)

Replacing the deterministic rounding of §4.4 with *single-shot randomized rounding* restores a tractable, everywhere-positive proposal density and yields an exactly reversible chain. One step of Chain E is as follows.

1. Pick an adjacent district pair (V_i, V_j) uniformly at random from the $|\mathrm{bp}(P)|$ boundary pairs of the current plan P;

2. Solve the (deterministic) QP of Definition 4.1 on H for $x^\star \in [0,1]^{|V_H|}$;

3. Draw the proposal $x' \in \{0,1\}^{|V_H|}$ by independent Bernoulli rounding from the *clipped* optimum, $x'_v \sim \mathrm{Bern}\big(\mathrm{clip}(x^\star_v; \delta, 1-\delta)\big)$ with $\delta = 10^{-12}$, *once*, with no retry on infeasibility. Clipping is part of the proposal's definition. The QP optimum can have active box bounds (coordinates exactly 0 or 1, e.g. at interior vertices pinned by the fidelity term), and sampling from the same clipped vector whose density enters the acceptance ratio is what makes the proposal law and its stated density identical, with every binary x' receiving positive probability;

4. Accept with probability
$$\min\Big(1,\ \frac{\pi(P')}{\pi(P)}\,\frac{q(x_0 \mid P', \{i,j\})}{q(x' \mid P, \{i,j\})}\,\frac{|\mathrm{bp}(P)|}{|\mathrm{bp}(P')|}\Big), \tag{9}$$
where $q(\cdot \mid P, \{i,j\})$ is the Bernoulli product density induced by the QP optimum at the corresponding state, and proposals that violate contiguity or the balance window (4) are rejected outright.

Here π is any target on plans; the paper uses $\pi_w(P) \propto \exp(-\lambda\, w(\partial P))$ for a kernel w, which reduces to the tempered cut-count target for the uniform kernel, and in §4.7 also the tree-weighted variant.

Proposition 4.3 (Detailed balance of Chain E)**.** *Chain E satisfies detailed balance with respect to any target π on $\mathcal{P}_k(G;\varepsilon)$ that is positive on valid plans.*

Throughout, fix the deterministic edge kernel w (weight jitter is never used in this chain) and write $\Omega = \mathcal{P}_k(G;\varepsilon)$ for the space of valid plans. Plans are *labeled* assignments in this subsection, so that a Bernoulli draw and its complement are distinct states and the proposal map is a bijection; on unlabeled partitions the map is two-to-one and the transition probability would need both terms. Since π depends only on the partition it is label-invariant, and forgetting labels after sampling recovers the unlabeled law. For a plan P and an adjacent pair $\{i,j\} \in \mathrm{bp}(P)$ write $H = V_i \cup V_j$ and let $x_0(P) \in \{0,1\}^{V_H}$ denote the indicator of V_i on H (the pair is drawn from a sorted list, so which district plays the role of i is a deterministic function of the unordered pair). Say that $P' \in \Omega$ is *pair-adjacent* to $P \in \Omega$ if $P' \neq P$, the two plans differ on exactly two districts $\{i,j\}$, and $H = V_i \cup V_j$ is connected. This relation is clearly symmetric in P and P'.

Lemma 4.2 (Proposal regularity)**.** *Let $P, P' \in \Omega$ be pair-adjacent via districts $\{i,j\}$, and suppose $\beta > 0$. Then:*

(i) $\{i,j\}$ is the unique unordered pair whose selection can generate the move $P \to P'$ or the move $P' \to P$, and it is selectable from both endpoints: $\{i,j\} \in \mathrm{bp}(P) \cap \mathrm{bp}(P')$.

(ii) The QP of Definition 4.1 on H with the global window (4) has the same (nonempty) feasible set for the forward and reverse solves, and its optimum is unique, hence a deterministic function of $(\text{endpoint}, \{i,j\})$.

(iii) The clipped-Bernoulli density $q(\cdot \mid \cdot, \{i,j\})$ is bounded below by $\delta^{|V_H|} > 0$ on every binary labeling of V_H; in particular $q(x_0(P') \mid P, \{i,j\}) > 0$ and $q(x_0(P) \mid P', \{i,j\}) > 0$.

Proof. (i) A step that selects pair $\{a,b\}$ relabels vertices only between districts a and b and changes no other district. If $P' \neq P$, at least one vertex of H switched between i and j, so districts i and j both changed; any generating pair must therefore equal $\{i,j\}$. For selectability, note that (V_i, V_j) and (V'_i, V'_j)

are both bipartitions of the connected graph H into nonempty parts (all four sets are nonempty districts of valid plans), and every such bipartition is crossed by at least one edge of H; hence the two districts are adjacent in both plans.

(ii) The feasible set is $\{x \in [0,1]^{V_H} : \ell \le p^\top x \le u\}$ with ℓ, u given by (4). It depends only on $(H, \varepsilon, \bar{p})$, not on the endpoint, so the forward and reverse solves optimize over the same set. It is nonempty at both endpoints because the window of (4) is precisely the condition that both sides of the split be valid districts, which $x_0(P)$ and $x_0(P')$ satisfy by validity of P and P'. This is where globality is essential. Anchoring the window to the merged-pair half $(1 \pm \varepsilon)\, p(V_H)/2$ instead lets a valid plan fail its own pair's window, zeroing a reverse density whose forward counterpart is positive and destroying reversibility outright. The objective $\alpha\, x^\top L_H x + \beta \,\|x - x_0\|^2$ has Hessian $2(\alpha L_H + \beta I) \succ 0$ for $\beta > 0$, so the minimizer over the (convex, compact, nonempty) feasible set exists and is unique, and it is a function of (L_H, x_0, ℓ, u) only. Since the kernel w is deterministic, L_H depends only on H, and x_0 only on the endpoint and the pair.

(iii) Clipping maps the optimum into $[\delta, 1-\delta]^{V_H}$, so every Bernoulli factor of the product density lies in $[\delta, 1-\delta]$ and every outcome in $\{0,1\}^{V_H}$ has probability at least $\delta^{|V_H|}$. □

Proof of Proposition 4.3. Let K denote the transition kernel of Chain E and fix distinct $P, P' \in \Omega$. If P and P' are not pair-adjacent, no single step can produce either move. A step changes at most the two selected districts, so a generating pair would have to be the differing pair; if the plans differ on more than two districts no pair qualifies, and if the union H of the two differing districts is disconnected, that pair is not adjacent in either plan and is never selected. Hence $K(P, P') = K(P', P) = 0$ and detailed balance holds trivially.

Otherwise let $\{i, j\}$ be the unique generating pair, abbreviate $q_\rightarrow = q(x_0(P') \mid P, \{i, j\})$ and $q_\leftarrow = q(x_0(P) \mid P', \{i, j\})$, and note that both are positive and finite by Lemma 4.2(ii)–(iii). Because the rounding is single-shot, $q_\rightarrow$ is the exact probability that the proposal from P with pair $\{i, j\}$ equals P'. No renormalization by a feasibility event is involved (a retry-until-feasible loop would replace q by a conditional law whose state-dependent normalizing constant does not cancel in the ratio), and clipping *before* sampling makes the sampled law and the density in (9) the same object. Hence

$$K(P, P') \;=\; \frac{1}{|\mathrm{bp}(P)|}\, q_\rightarrow\, \alpha(P, P'), \qquad \alpha(P, P') = \min\Big(1,\; \frac{\pi(P')}{\pi(P)}\, \frac{q_\leftarrow}{q_\rightarrow}\, \frac{|\mathrm{bp}(P)|}{|\mathrm{bp}(P')|}\Big).$$

Multiplying through,

$$\pi(P)\, K(P, P') \;=\; \min\Big(\frac{\pi(P)\, q_\rightarrow}{|\mathrm{bp}(P)|},\; \frac{\pi(P')\, q_\leftarrow}{|\mathrm{bp}(P')|}\Big),$$

an expression symmetric under exchanging $(P, q_\rightarrow)$ with $(P', q_\leftarrow)$. The identical computation started from P' gives the same right-hand side, so $\pi(P)K(P, P') = \pi(P')K(P', P)$.

It remains to check that the rejection pathways do not disturb this identity. A Bernoulli draw whose split leaves a side empty, disconnected, or outside the window (4) is a proposal onto a state of zero target mass and is rejected with probability one, contributing only to the diagonal $K(P, P)$, for which detailed balance is vacuous; the same applies to in-space proposals rejected by (9). Rejection on solver failure is symmetric because, by Lemma 4.2(ii), the solve is a deterministic function of $(H, \{i, j\}, \text{endpoint})$, so a failure that suppresses the move $P \to P'$ suppresses the reverse move identically. Detailed balance of K with respect to π on Ω follows, and with it stationarity of π [51]. □

Remark 4.1 (Scope, verification, and ergodicity)**.** *The proof treats the QP solve as exact; in floating point the minimizer is computed to solver tolerance, and determinism of the implementation (fixed kernel, no jitter, sorted pair enumeration) is what the argument actually uses. As an end-to-end check, we assembled the full transition kernel of the implementation on an exhaustively enumerated* 2×3 *grid* ($k = 2$, *tolerance* $\varepsilon = 1/3$, *so that the 18 labeled valid states include* 4+2 *as well as* 3+3 *splits) and*

verified $\pi(P)K(P,P') = \pi(P')K(P',P)$ *entrywise, with maximum relative violation* 1.5×10^{-15} *(machine precision;* `db_check_2x3.py`*). Detailed balance gives stationarity, not ergodicity. Within a selected pair every binary relabeling of* H *has positive proposal probability by Lemma 4.2(iii), but irreducibility of Chain E across pairs on general* (G, k, ε) *is not established, so convergence statements are for the chain restricted to its communicating class. On the enumerated* 5×5 *fiber the chain's cut distribution is indistinguishable from its target (§5.1), on the* 2×3 *fiber a million-step run's full occupancy distribution is (§5.2), and the principled route to global mobility is the mixture of §4.7.*

Chain E trades mobility for exactness. Because a proposal must be feasible in a single Bernoulli draw, and because Corollary 4.1 caps how far a draw can reach, the chain behaves as a strictly local sampler. §4.6 measures exactly why, and the answer is feasibility rather than the Metropolis ratio. We therefore present Chain E as the local component of the sampler rather than as a practical standalone sampler. Mixing E with a global kernel preserves π only if that kernel is *also* π-invariant. RevReCom as published is reversible with respect to its spanning-tree measure, not an arbitrary π, so a naive mixture would preserve neither; the correct construction Metropolizes the tree move toward π using its computable Hastings ratio [11, 4] and mixes *that* kernel with E, which is the construction of §4.7.

On the tempered baseline (Chain A). Our Tempered ReCom baseline applies the filter $\min(1, \exp(-\lambda\Delta|\partial P|))$ over the ReCom proposal, which *assumes a symmetric proposal.* ReCom's proposal is not symmetric; the true Hastings ratio involves spanning-tree counts of the affected districts [4, 11], so Chain A targets π only approximately, and §5.1 quantifies the bias on enumerated ground truth. We keep the symmetric-q baseline because it is what practitioners run.

4.6 Mobility of the exact chain: feasibility, not acceptance

Proposition 4.2 says how far a proposal reaches. Mobility is that reach times the probability the proposal is kept, and the second factor has two causes of different kinds. A Bernoulli draw can produce a split that is disconnected or outside the balance window, in which case it is rejected with probability one before any coin is flipped. Or it can be a valid plan that loses the Metropolis coin. We measured both on the enumerated 5×5 fiber, with $\alpha = \tau$ and $\beta = 1$, over 3 000 steps per timestep from the same initial plan (Table 2).

τ	$\mathbb{E}$[#flips]	single-flip share	feasible | single	Pr[feasible]	accept | feasible	units moved / step	states
0.01	0.036	0.035	0.000	0.964	1.000	0.0000	1
0.03	0.126	0.110	0.000	0.884	1.000	0.0033	6
0.10	0.385	0.268	0.000	0.683	0.999	0.0093	12
0.30	0.909	0.349	0.000	0.437	0.989	**0.0353**	**38**
1.00	1.914	0.254	0.000	0.193	0.983	0.0313	37
3.00	2.672	0.168	0.000	0.074	0.986	0.0187	19

Table 2: Chain E against the diffusion timestep on the 5×5 fiber ($k = 5$, exact balance). "Feasible | single" is the feasibility rate of single-unit proposals, "accept | feasible" the Metropolis acceptance rate among feasible proposals, and "units moved / step" the product of acceptance and accepted move size, the quantity a tuning rule should maximize. "States" counts distinct fiber states reached in 3 000 steps. The locked setting of Appendix A is $\tau = 0.10$.

Three facts follow. First, every single-unit proposal is infeasible at every timestep, exactly as exact balance requires, since reassigning one cell produces districts of sizes 4 and 6. The leading-order move of

the diffusion proposal is therefore always rejected on this fiber, and the chain moves only through second-order coordinated draws. This is the mechanism behind the coverage figures of §5.1, derived rather than observed. Second, the Metropolis ratio is never the bottleneck. Given a feasible proposal, acceptance is between 98 and 100 percent throughout. The chain's stickiness is a feasibility effect, and the natural phrase "large moves are rarely accepted" is wrong in an instructive way. They are rarely feasible, and when feasible they are accepted. Third, acceptance rate is a misleading diagnostic for this chain. At $\tau = 0.01$ it is 96 percent while the chain never leaves its initial state, because 96 percent of proposals flip nothing. We report units moved per step instead.

Units moved per step has an interior maximum near $\tau = 0.3$. The setting locked in Appendix A, $\tau = 0.1$, sits about 3.8× below it, so on this fiber the chain can be improved for free; §5.3 shows the optimum does not transfer to Iowa, where the balance window binds more often at larger τ, and §5.5 shows that on large pairs the dial has to be turned the other way. Even at the optimum a thousand steps move about 35 units, and Corollary 4.1 says no timestep can lift the per-proposal reach above half the merged pair. Both statements matter. Chain E can be several times more mobile than shown, and several times is not enough, which is why a global move is required (§4.7). Under a loose tolerance single flips become feasible and mobility gains a first-order term in τ, which is why the same chain climbs steadily on Iowa (§5.6) while freezing on the exact-balance grid.

The mobility law. On an exact-balance fiber the cheapest feasible draw is a swap, one unit $u \in V_i$ and one unit $v \in V_j$ changing sides together, which preserves both populations. Rounding flips units independently with probabilities $p_v = |x^\star_v - x_{0,v}|$, so the probability that exactly $\{u, v\}$ flips is $p_u p_v \prod_{w \neq u,v}(1 - p_w)$, and the two-swap channel contributes

$$M_2 \;=\; 2 \sum_{\substack{u \in V_i,\ v \in V_j \\ \text{swap feasible}}} p_u\, p_v \prod_{w \neq u,v} (1 - p_w)\; \alpha(u, v) \tag{10}$$

units per step, where the sum runs over cross-boundary pairs whose swap keeps both districts connected and $\alpha(u, v)$ is the Metropolis probability of that swap, each of which costs one reverse linear solve. For small τ, $p_v = \tau b_v + O(\tau^2)$ with b_v the kernel weight of v's cross-boundary edges, so $M_2 = 2\tau^2 S_2 + O(\tau^3)$ with $S_2 = \sum_{\text{feasible swaps}} b_u b_v$, a coefficient computable from the current plan. We checked the law by running Chain E for 20 000 steps per timestep and evaluating (10) at every step for the pair chosen and the plan visited (Table 3).

τ	measured units / step	M_2	M_2 without α	$2\tau^2 S_2$
0.03	0.0021	0.0018	0.0024	0.0033
0.10	0.0105	0.0105	0.0127	0.0341
0.30	0.0320	0.0326	0.0383	0.3628

Table 3: The mobility law against Chain E on the 5 × 5 fiber, 20 000 steps per timestep. The two-swap channel with its Metropolis factor matches the measured mobility to within counting noise at every timestep; the $\tau = 0.03$ run holds 21 accepted swaps.

The two-swap channel accounts for the chain's mobility at every timestep tested, and four-unit draws contribute nothing measurable even at $\tau = 0.3$. The Metropolis factor is not optional. Without it the channel overpredicts by about 20%, because the chain's near-unit acceptance given feasibility is dominated by zero-flip proposals, and a genuine swap that lengthens the boundary is accepted less often. The small-timestep form is accurate only for $\tau \lesssim 0.03$, a narrower range than the move-size law's, as expected for a second-order quantity. Under a loose tolerance single flips become feasible, the law gains a first-order channel, and the same calculation applies with single units in place of swaps.

4.7 Exact tree moves, the lifted local move, and the reversible hybrid

Corollary 4.1 says the local chain cannot be made mobile by tuning, so a global move is needed, and §4.5 says that move must be reversible for the same target. This subsection gives two exact tree moves, the one we use and the one that explains Chain A, lifts Chain E to the state space of the first, and states the sampler. It ends with the fact that organizes the experiments: an exact tree move proposes from one fixed law, and pays for every target that differs from it.

4.7.1 The partition-level tree move (Chain T)

Pick an adjacent pair $\{i, j\}$ uniformly from $\mathrm{bp}(P)$, draw a uniform spanning tree T of $H = G[V_i \cup V_j]$ by Wilson's algorithm, and pick one edge of T uniformly among *all* $|V_H| - 1$ tree edges. If the two components A, B both lie in the window (4), assign the labels i, j to them by a fair coin and propose; otherwise reject. Every spanning tree of H that yields $\{A, B\}$ on removal of one edge is a spanning tree of $G[A]$, a spanning tree of $G[B]$, and one crossing edge, so

$$q(P \to P') \;=\; \frac{\tau(A)\,\tau(B)\,|\partial(A,B)|}{2\,|\mathrm{bp}(P)|\,\tau(H)\,(|V_H|-1)},$$

where $\tau(\cdot)$ is the spanning-tree count by the Matrix–Tree theorem and $|\partial(A,B)|$ the number of edges between A and B. The reverse move shares H, so $\tau(H)$ and $|V_H| - 1$ cancel and the move is accepted with probability

$$\min\Big(1,\; \frac{\pi(P')}{\pi(P)}\;\frac{\tau(V_i)\,\tau(V_j)\,|\partial(V_i,V_j)|}{\tau(A)\,\tau(B)\,|\partial(A,B)|}\;\frac{|\mathrm{bp}(P)|}{|\mathrm{bp}(P')|}\Big),$$

computable from four determinants and two edge counts. The fair coin makes the move a bijection on labeled plans. The cut edge is drawn among all tree edges rather than among balanced ones, and that is the point of this move for us. GerryChain's `recom` draws among balanced edges, which multiplies the density by a tree-dependent normalization that does not cancel at the partition level, and that is precisely why Chain A is only approximately correct. The price of the closed form is acceptance. Most tree edges split a merged pair unevenly, so the move accepts about 7% of proposals on the exact-balance 5×5 fiber and about 1% on Iowa at $\varepsilon = 0.05$. It is not what practice runs, and an earlier version of this work used it as the baseline for the hybrid; the numbers in §5.3 show why that was the wrong comparison.

4.7.2 The forest move (Chain F)

Recovering balanced-edge selection with an exact density means lifting the state to a spanning forest, as Metropolized forest recombination does [4]. A labeled forest is $\mathcal{F} = (P, \{T_d\})$, where T_d is a spanning tree of $G[V_d]$ for every district. The fiber over a plan P has $\tau(P) = \prod_d \tau(V_d)$ forests, and the lifted target

$$\tilde{\pi}(\mathcal{F}) \;=\; \frac{\pi(P(\mathcal{F}))}{\tau(P(\mathcal{F}))} \tag{11}$$

has marginal π on plans. Every statistic we report is a function of the plan, so traces are comparable with the partition-level chains.

One step. Pick $\{i, j\}$ uniformly from $\mathrm{bp}(P)$ and draw a uniform spanning tree T of H. Let $B(T)$ be the set of edges of T whose removal leaves both components inside the window (4). If $B(T)$ is empty, draw again, up to M times, then reject. Otherwise pick $e \in B(T)$ uniformly, let $\{A, B\}$ be the components, and assign the labels by a fair coin. A forest $\mathcal{F}' = (\ldots, T_A, T_B, \ldots)$ is reached from exactly those draws $(\hat{T}, \hat{e})$ with $\hat{T} = T_A \cup T_B \cup f$ for a crossing edge $f \in \partial(A, B)$ and $\hat{e} = f$, so

$$q(\mathcal{F} \to \mathcal{F}') \;=\; \frac{c_H}{2\,|\mathrm{bp}(P)|\,\tau(H)}\,S(T_A,T_B), \qquad S(T_A,T_B) \;=\; \sum_{f \in \partial(A,B)} \frac{1}{|B(T_A \cup T_B \cup f)|}, \tag{12}$$

where $c_H = (1-(1-p_H)^M)/p_H$ accounts for the retry, p_H being the probability that a uniform spanning tree of H has a balanced edge. The retry is exact because p_H depends on H and the window only, never on the endpoint, and the reverse move shares H, so c_H and $\tau(H)$ cancel. (This is not true of the local move's rounding, whose feasibility event depends on the QP optimum at the endpoint; that is why Chain E must be single-shot.) The Hastings ratio is

$$\frac{\pi(P')}{\pi(P)} \cdot \frac{\tau(V_i)\,\tau(V_j)}{\tau(A)\,\tau(B)} \cdot \frac{S(T_i,T_j)}{S(T_A,T_B)} \cdot \frac{|\mathrm{bp}(P)|}{|\mathrm{bp}(P')|}, \tag{13}$$

where the tree-count ratio is the lifted target's, not the proposal's. Each S is a sum over the crossing edges of an $O(n)$ subtree-population pass, and the two determinants of the current districts are cached, so a step costs two determinants plus $O(|\partial|\,n)$. Our implementation differs from Autry et al. in one convention: the adjacent pair is drawn uniformly from boundary pairs, as in every other move of this paper, rather than by a uniform cut edge; both are exact and the ratio carries the corresponding factor.

Two targets. Besides $\pi_w(P) \propto e^{-\lambda\, w(\partial P)}$ we also use its tree-weighted version $\pi_w^{\mathrm{tree}}(P) \propto \tau(P)\, e^{-\lambda\, w(\partial P)}$, the measure that §5.1 identifies as close to what tempered ReCom samples. Under π_w^{tree} the tree-count ratio in (13) cancels against the same factor in $\pi(P')/\pi(P)$. Both targets are handled by the same code path; the local move's acceptance under the tree-weighted target acquires the tree-count ratio instead, since it changes the plan and hence $\tau(P)$.

4.7.3 The lifted local move and the hybrid (Chain H)

Run Chain E on the plan against the chosen target. If it accepts, redraw the trees of the two changed districts uniformly by Wilson's algorithm; otherwise keep the forest. With K_E π-reversible on plans, the lifted kernel is $\tilde{K}(\mathcal{F},\mathcal{F}') = K_E(P,P')/(\tau(A)\tau(B))$, and $\tilde{\pi}(\mathcal{F})\tilde{K}(\mathcal{F},\mathcal{F}')$ is symmetric because $\tau(P) = \tau(V_i)\tau(V_j)$ times the other districts' counts, which the move does not change. Where the balance window is slack the local solve is the linear solve of Lemma 4.1, which we use in place of the QP solver; both paths are deterministic functions of the state and pair, which is all Proposition 4.3 requires.

The hybrid takes a lifted local step with probability p_{loc} and a forest step otherwise. Both kernels are $\tilde{\pi}$-reversible, so the mixture is, and its marginal on plans is π.

Proposition 4.4 (Ergodicity of the hybrid)**.** *If the forest move is irreducible on the lifted space (which holds under the standard assumptions for recombination moves; see [22, 9, 4]), then the mixture with the lifted local move at any $p_{\mathrm{loc}} \in (0,1)$ is ergodic.*

Proof. Write $K = (1-p_{\mathrm{loc}})K_{\mathrm{F}} + p_{\mathrm{loc}}\tilde{K}_{\mathrm{E}}$. If $K_{\mathrm{F}}^t(\mathcal{F},\mathcal{F}') > 0$ for some t, then $K^t(\mathcal{F},\mathcal{F}') \geq (1-p_{\mathrm{loc}})^t\, K_{\mathrm{F}}^t(\mathcal{F},\mathcal{F}') > 0$, so irreducibility of the forest move implies irreducibility of the mixture. The forest move rejects with positive probability at every state, so $K(\mathcal{F},\mathcal{F}) > 0$ and the chain is aperiodic. □

Verification. On the lifted 2×3 fiber (18 labeled plans, 30 forests) we assembled the forest kernel, the lifted local kernel with the linear-solve path, and their mixture analytically and checked $\tilde{\pi}(\mathcal{F})K(\mathcal{F},\mathcal{F}') = \tilde{\pi}(\mathcal{F}')K(\mathcal{F}',\mathcal{F})$ entrywise. The maximum relative violation is 2.4×10^{-16} for the forest move, 1.4×10^{-15} for the lifted local move, and 2.6×10^{-16} for the mixture under the uniform kernel at $\lambda = 0.1$, and 4.1×10^{-16}, 1.1×10^{-15} and 6.0×10^{-16} under a density kernel at $\lambda = 1$, for both targets. The closed-form proposal mass (12) equals the mass obtained by enumerating every (tree, balanced edge, coin) draw to 1.8×10^{-16} (`forest_recom.py`). The same check for the partition-level chains T, E and their mixture gives 4.8×10^{-16}, 1.6×10^{-15} and 7.2×10^{-16} (`hybrid.py`). §5.3 tests the forest move and the hybrid against enumerated targets on 4 006 plans.

4.7.4 What an exact tree move pays

Both tree moves draw the proposed split from the same law, the spanning-tree measure on balanced bipartitions of H conditioned on nothing else: Chain T from it exactly, Chain F from it up to the S factor, which the experiments below show to be within a factor of two of one. Given the pair, the proposed split does not depend on the current one. On a merged pair the tree move is therefore an *independence sampler*, and its acceptance at equilibrium is governed by how far the target's conditional on the pair sits from that law. Two terms of (13) carry the distance. Under π_w the tree-count ratio does: a uniform spanning tree favors splits with many spanning trees, and by Procaccia and Tucker-Foltz [44] the logarithm of a tree count scales with the boundary length, so at the equilibrium of a target that is less compact than the tree measure the ratio is typically far below one. Under π_w^{tree} that ratio cancels and the target ratio $e^{-\lambda\Delta w(\partial)}$ carries the distance instead, which is large whenever the tempering makes the target more compact than an untempered tree split.

We measured the terms on Iowa over 3 000 forest proposals from a tree-partition start under π_w (Table 4). The tree-count term dominates, with mean logarithm -6 to -10 and a share of 67 to 84 percent of feasible proposals below $e^{-2.3}$ on that term alone, while the target, S and pair-count terms are mild. At the equilibrium of the uniform target the acceptance given a feasible tree is 0.17; under the tree-weighted target at the same temperature it is 0.74, because at λ equivalent to 0.1 that target is close to the untempered tree measure. The scaling experiment of §5.5 then shows what happens as the pair grows: for a compactness-tempered target, with or without tree weights, the mismatch grows in proportion to the boundary length and the acceptance decays exponentially in the grid side. The local move loses acceptance with pair size as well, because its expected move grows with the boundary (Corollary 4.1), but it has a dial for it, the timestep; the tree move has none, since its proposal is the same law whatever the target.

target	Pr[balanced]	acc. | feas.	mean of each log term				share, τ-term $< e^{-2.3}$
			target	tree count	S	bp	
uniform, $\lambda = 0.10$	0.41	0.25	+0.63	−5.93	+0.47	−0.01	0.67
perimeter, $\lambda = 0.34$	0.53	0.22	+0.65	−5.98	+0.59	−0.00	0.69
density, $\lambda = 0.72$	0.41	0.13	+1.62	−9.78	+0.74	−0.06	0.84

Table 4: The four terms of the forest move's log Hastings ratio (13) on Iowa ($k = 4$, $\varepsilon = 0.05$) under π_w, averaged over the feasible proposals of 3 000 steps from a tree-partition start, without retry. The tree-count ratio is the rejection; the retry raises Pr[tree balanced] to 0.95 at 2.5 times the wall clock and leaves accepted moves per second unchanged.

5 Experiments

We evaluate the chains of Table 1 on the 5×5 grid (full enumeration, ground-truth target), the 2×3 grid (18 states, full convergence checkable), the Iowa county dual graph (real geography, 99 counties, $k = 4$, $\varepsilon = 0.05$), and $N \times N$ grids with $k = 2$ for the scaling experiment. All hyperparameters are locked in Appendix A; raw traces, per-step balance audits, and configuration stamps are released in the project repository.

Statistical protocol. Chains are autocorrelated, so we do not report i.i.d. test statistics on raw traces. Effective sample sizes (ESS) come from Geyer's initial-positive-sequence estimator of the integrated autocorrelation time, computed by FFT, and are appropriate for inference about cut means only; distributional

tests use batch means (§5.2), and distances to enumerated targets are read against parametric-bootstrap noise floors at the ESS (§5.1). When two chains are compared by effective samples per second we also report effective samples per step, which does not depend on the implementation, and we check that the two chains sit at the same level, by quarter means and by the difference of their post-burn means in standard errors from their own ESS. Each chain runs on its own seeded RNG stream; the baseline tables report seed 42, and the ten-seed means of Chains V and A (seeds 42 through 51) are quoted where they enter the calibration. Between-seed standard deviations of the baselines are about 0.4 cuts on Iowa and 0.1 to 0.2 cuts on the 5×5 fiber, so single-seed means are representative.

Balance audit. Every run of the baseline chains records, at every step, the maximum relative deviation of any district population from the global ideal, and all of them satisfy the $\varepsilon = 0.05$ bound at every step (Iowa maxima 0.0498–0.0500 for the runs of §5.6); the exact chains of §4.7 enforce the window inside each move by construction, rejecting any draw outside it. We flag this because it is not automatic. GerryChain's raw `recom` proposal validates only one side of each tree cut, and an earlier version of our own local steps anchored balance to the merged-pair half rather than the global ideal (§4.2); either slip lets district populations drift to $\sim 4\times$ the nominal tolerance on Iowa before the audit catches it. We recommend per-step balance auditing as standard practice for any custom chain harness that bypasses GerryChain's validator pipeline.

5.1 Ground truth on the enumerated fiber: what the family samples

The 5×5 partition fiber (the set of all valid plans; $k = 5$, exact balance, 4 006 partitions; Appendix B) admits full enumeration, so every distributional quantity is exactly computable. The chains run with $\varepsilon = 0.05$, which on 25 unit-population cells admits only districts of exactly five cells, so the state space the chains traverse is precisely this enumerated fiber. Uniform-on-fiber is not the target of any chain here; the total-variation column is a descriptive concentration measure only, and with about 900 correlated draws over 4 006 states even an independent uniform sampler would show TV above 0.78. The informative comparisons are the cut means against the exact target mean and the long runs of §5.1, where distances are read against proper noise floors. Table 5 gives the short-run picture.

Chain	Mean cuts	Std	Unique / 4006	TV(emp, uniform)
V: Vanilla ReCom	17.06	1.08	337 / 4006 (8.4%)	0.92
A: Tempered ReCom	17.02	1.15	311 / 4006 (7.8%)	0.92
E: QP-MH exact	16.27	0.44	6 / 4006 (0.1%)	1.00

Table 5: 5×5 short-run comparison ($k = 5$, exact balance, 1 000 steps, burn-in 100, seed 42). The exact target mean is $\mathbb{E}_\pi[|\partial P|] = 18.56$ at $\lambda = 0.10$, and *every* chain sits far below it at this horizon, including the exactly reversible Chain E, which at 1 000 steps has visited only 6 states. Exactness of the kernel does not buy mixing speed.

The origin of the gap. The gap between every chain's empirical mean and the target mean 18.56 quantifies the compactness bias *shared* by the whole ReCom-derived family, and on the enumerated fiber we can identify its origin. Using the Matrix–Tree theorem we compute, for each of the 4 006 partitions, the spanning-tree-count weight $\prod_i \tau(V_i)$, the measure that the RevReCom analysis identifies as the natural target of tree-based recombination [11, 4] and whose relation to compactness Procaccia and Tucker-Foltz [44] proved. At the level of means the match is close. The tree-weighted measure has $\mathbb{E}[|\partial P|] = 16.88$ against Vanilla ReCom's ten-seed mean of 16.95 ± 0.11, and the tree-weighted-and-tempered measure $\prod_i \tau(V_i)\, e^{-\lambda|\partial P|}$ has $\mathbb{E}[|\partial P|] = 16.78$ against Chain A's ten-seed mean of 16.71 ± 0.16;

both gaps are 0.07 cuts, within one between-seed standard deviation, and the two quasi-targets sit 1.7 and 1.8 cuts below the nominal tempered mean. A mean is a one-dimensional summary, though, and many distributions share one. The long runs below compare full distributions, which is where the claim is actually tested.

Long-run validation at the level of distributions. We run Chain A for 2×10^5 steps and Chain E for 10^5 steps on the enumerated fiber (seed 42, 10% burn-in) and compare each chain's empirical distribution against three exactly computable reference measures, the nominal target $\pi \propto e^{-\lambda|\partial P|}$, the tree-weighted measure $\prod_i \tau(V_i)$, and the tree-weighted-and-tempered measure. Two design choices make the comparison informative. First, a raw total-variation distance is uninterpretable on 4 006 states, since even an independent sampler shows large TV until the draw count greatly exceeds the state count. So every distance is reported against a parametric-bootstrap noise floor, the same distance computed for independent draws from the reference measure at the chain's effective sample size. A converged chain sits at its floor; a chain converging elsewhere sits above it by many floor standard deviations. Second, the headline statistic is the cut-count marginal rather than the full state distribution. The fiber's cut counts take five values, so with 1.8×10^5 retained draws the marginal is estimated to three decimals and its floor is small, which removes the coverage problem entirely. Table 6 and Figure 3 give the result.

Chain	Reference measure	TV	Floor at ESS	z
A (2×10^5 steps, ESS 12 312)	nominal π	0.629	0.007	+220
	tree-weighted	0.065	0.006	+22
	tree-weighted, tempered	0.022	0.006	+6.7
E (10^5 steps, ESS 28)	nominal π	0.079	0.141	−1.2
	tree-weighted	0.493	0.125	+6.9
	tree-weighted, tempered	0.532	0.125	+7.5

Table 6: Total variation between each chain's empirical cut-count marginal and three reference measures on the enumerated 5×5 fiber, with the parametric-bootstrap noise floor at the chain's effective sample size and the excess in floor standard deviations. Chain A is far from its nominal target and close to, though distinguishable from, the tree-weighted tempered measure. Chain E is indistinguishable from the nominal target it is proved reversible for and far from the tree measures.

Chain A does not sample its nominal target, and the gap is a plateau, not a drift. At 2×10^5 steps the tempered chain has visited 3 524/4 006 states, its mean is 16.73, and its cut distribution sits 220 floor standard deviations from the nominal target, at TV 0.63. Figure 3(b) shows that distance flat across five decades of run length while the distance to the tree-weighted tempered measure falls toward the floor. This is convergence to a different distribution, not slow mixing, and it is what the symmetric-q filter of §4.5 predicts. The match to the tree-weighted tempered measure is close but not exact. At TV 0.022 the chain sits 6.7 floor standard deviations above it, so the tree measure is a very good description of what tempered ReCom samples and not a perfect one, which is expected, since Vanilla ReCom's stationary law is not exactly the tree measure and RevReCom exists to make it so. On the full 4 006-state distribution the noise floor at effective sample size is 0.15 to 0.23, so that comparison can separate the nominal target ($z = +164$) but not the two tree measures from each other, which is why the marginal carries the claim.

Chain E samples what it states. At 10^5 steps its cut marginal is 1.2 floor standard deviations from the nominal target and about 7 from either tree measure. The two chains separate exactly as theory predicts. Convergence of the full distribution is far slower for this sticky local chain, as §4.6 explains. It has visited 403/4 006 states, and its full-state TV of 0.90 sits at a floor of 0.99 at effective sample size 28, so that test has no power here. The fiber on which full convergence can be checked is the next subsection.

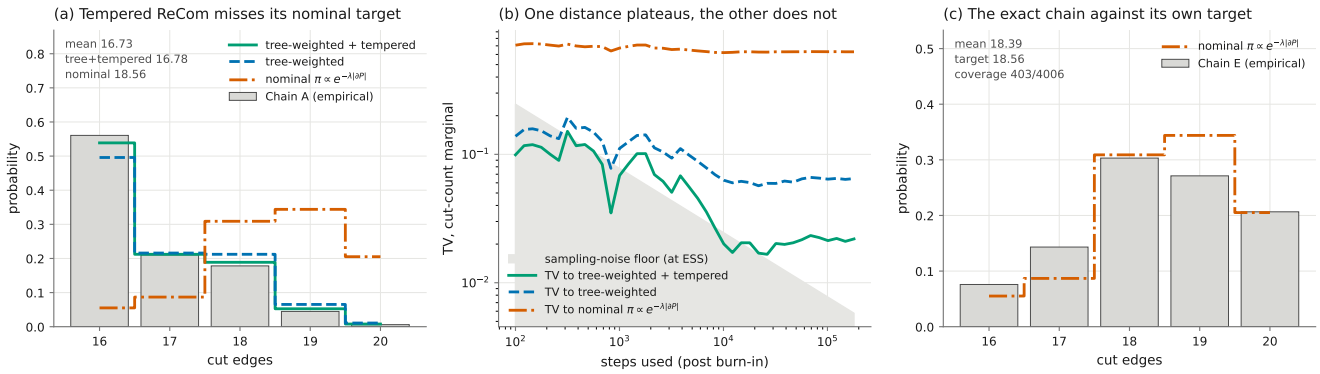


Figure 3: What the chains sample. (a) Chain A's cut-count marginal after 2×10^5 steps against the three reference measures. (b) Total variation of the running marginal to each reference as a function of run length, with the noise floor at the effective sample size shaded; the distance to the nominal target is a plateau across five decades while the distance to the tree-weighted tempered measure descends toward the floor. (c) Chain E's marginal after 10^5 steps against its own target.

5.2 Exact convergence on the 18-state fiber

Detailed balance gives stationarity, not convergence, and on 4 006 states the exact chain cannot be run to convergence in reasonable time. On the 2×3 grid with $k = 2$ and tolerance $\varepsilon = 1/3$, the fiber of labeled valid plans has 18 states, the same state space on which Remark 4.1 verifies the kernel to machine precision, and a long run can cover it. We run Chain E for 10^6 steps (seed 42, 10% burn-in, locked parameters $\alpha = 1$, $\beta = 10$, $\lambda = 0.10$). It visits all 18 states, its acceptance rate is 0.785, and its post-burn-in mean cut count is 2.75992 against the exact 2.76002.

The distributional test uses batch means, because thinning a chain by an effective sample size computed from a summary statistic is not valid here. The cut count takes two values on this fiber, so its integrated autocorrelation time (17 steps) understates that of the 18-state occupancy process (25 steps), and a chi-square test on a chain thinned by the former rejects spuriously. We therefore split the trace into 200 blocks of 4 500 steps, over 180 occupancy correlation times each, estimate the covariance of the occupancy vector from the between-block scatter, and test with Hotelling's T^2 on the 17 free coordinates. The result is $F(17, 183) = 0.87$, $p = 0.61$. The occupancy distribution is not distinguishable from the exact target. Total variation to the target is 0.0110, against a noise floor of about 0.009 at the occupancy effective sample size, and it fell from 0.0250 at 2×10^5 steps, a factor of 2.27 for five times the steps, which is the $\sqrt{n}$ rate of a converged chain. At 2×10^5 steps the same test gave $p = 0.07$ with three states in one cut-level set persistently under-visited; at 10^6 steps those deviations dissolved into the few $|z| \approx 2$ values that 18 states produce by chance. Together with the kernel check, this is end-to-end validation of the construction of §4.5. The implemented kernel satisfies detailed balance exactly, and the chain it generates reaches its target.

5.3 The forest move and the hybrid against enumerated targets

We run the partition-level tree move (T), the forest move (F) and the hybrid (H, $p_{\text{loc}} = 0.2$, $\tau = 0.1$) from the same start for 2×10^5 steps on the 5×5 fiber against three enumerated targets, the uniform kernel at $\lambda = 0.1$ and at $\lambda = 1$ and a density kernel at $\lambda = 1$ built from a synthetic central city, $\rho(x, y) = 3\exp(-r^2/2)$ with $w_{uv} = \exp(-(\rho_u - \rho_v)^2)$, under which π_w favors boundaries that follow the density gradient and the tree proposal is blind to the kernel; and then the forest move and the hybrid against the tree-weighted versions of the same three targets. Each chain is tested as in §5.1, by the

cut-count marginal against its noise floor and by batch-means Hotelling (with cut bins of negligible target mass merged, since under the sharp tree-weighted target the top bin has mass 3×10^{-4}), and we report effective samples per second. The pre-registered criterion for the local move is the ratio of the hybrid's effective samples per second to the forest move's: at or above 1.5 the local move earns its place, below 1 it is a passenger.

Target	Chain	TV marg	z	Hotel. p	acc global	acc local	ESS/s	ratio
target π_w								
uniform, λ=0.1	T	0.010	−1.8	0.58	0.066	—	51.1	
	F	0.006	−1.2	0.46	0.851	—	211.8	F/T 4.15
	H	0.013	+0.2	0.43	0.851	0.660	143.8	H/F 0.68
uniform, λ=1	T	0.028	−0.8	0.21	0.074	—	45.2	
	F	0.004	−1.3	0.80	0.979	—	338.6	F/T 7.49
	H	0.005	−1.3	0.83	0.979	0.673	240.3	H/F 0.71
density, λ=1	T	0.028	−0.6	0.60	0.069	—	55.3	
	F	0.004	−1.5	0.96	0.876	—	254.5	F/T 4.61
	H	0.005	−1.5	0.25	0.876	0.784	168.9	H/F 0.66
target π_w^{tree}								
uniform, λ=0.1	F	0.009	+0.2	0.37	0.944	—	257.3	
	H	0.006	−0.8	0.78	0.942	0.675	188.8	H/F 0.73
uniform, λ=1	F	0.003	−1.0	0.64	0.884	—	265.9	
	H	0.005	−0.4	0.41	0.886	0.683	182.3	H/F 0.69
density, λ=1	F	0.007	+0.1	0.46	0.906	—	259.3	
	H	0.003	−1.3	0.70	0.906	0.789	168.6	H/F 0.65

Table 7: Exact tree moves and the hybrid on the enumerated 5×5 fiber, 2×10^5 steps each, seed 42. "TV marg" is total variation of the cut-count marginal to the enumerated target and z its excess over the noise floor at the effective sample size; "Hotel. p" is the batch-means test of §5.2; ESS is of the weighted-cut trace, timed in the same session.

Three findings. First, every chain samples its target. All fifteen rows sit within 1.8 floor standard deviations of the enumerated marginal with Hotelling p between 0.21 and 0.96, under both target families and both kernels, so the forest move, the lifted local move and their mixture are correct for kernel-weighted targets on 4 006 states and not only on the 30 lifted states of the analytic check. Second, the forest move is a different baseline from the partition-level tree move: it accepts 0.85 to 0.98 against 0.07, and delivers 4 to 7.5 times the effective samples per second on this fiber. Third, on this fiber the local move is a passenger under every target, with ratios 0.65 to 0.73. That is what §4.6 predicts under exact balance, where the local move's first-order channel is closed, and it is also what §4.7.4 predicts for pairs of ten cells, where the mismatch between any of these targets and the tree measure is small. The grid tells us the construction is right and tells us nothing yet about where the local move matters; Iowa and the scaling experiment do.

5.4 The kill test on Iowa

On Iowa ($k = 4$, $\varepsilon = 0.05$) there is no ground truth, so the comparison is mobility at equilibrium. We use the paper's three kernels with λ scaled so that $\lambda\, w(\partial P)$ at the initial plan equals the uniform case's $0.1\,|\partial P|$, and run the forest move alone and the hybrid at $p_{\text{loc}} \in \{0.1, 0.3\}$, $\tau = 0.1$. The criterion was fixed before the long runs were started: under π_w, with $p_{\text{loc}} = 0.1$, the hybrid passes if its effective samples per second are at least 1.5 times the forest move's on both the perimeter and the density target

on both seeds, with the two chains at the same level. The $p_{\text{loc}} = 0.3$ runs were added after the short runs favored that setting and are reported as a secondary finding.

Short runs, direction and timing. Table 8 gives 2×10^5 steps per chain. Under π_w the forest move accepts 0.16 to 0.17, the partition-level tree move 0.006 to 0.012, and yet the two deliver similar effective samples per second, because a forest step costs six times more (every feasible draw pays the determinants and the crossing-edge sums) and the partition-level move pays them only on the one percent of draws that balance. The hybrid beats the forest move on every kernel, by 1.4 to 1.8 at $p_{\text{loc}} = 0.1$ and 2.4 to 4.4 at 0.3, but the forest move's effective sample size in these runs is 36 to 56, so these ratios carry roughly 40 percent uncertainty. Under π_w^{tree} the picture inverts: the forest move accepts 0.66 to 0.74, its effective sample size is about 10 000 per 2×10^5 steps, two hundred times its value under π_w at the same step count, and the hybrid is a passenger at 0.86 to 0.93.

Target	Chain	acc global	acc local	$\bar{w}$ 1st half	$\bar{w}$ 2nd half	ESS	ESS/s	ratio
π_w								
uniform, λ=0.10	F	0.163	—	89.2	87.8	36	0.161	—
	H, p=0.1	0.158	0.249	90.1	89.4	56	0.223	1.39
	H, p=0.3	0.165	0.254	89.8	89.3	180	0.714	4.44
	T	0.012	—	83.2	87.2	14	0.395	2.46
perimeter, λ=0.34	F	0.163	—	28.1	28.2	43	0.185	—
	H, p=0.1	0.163	0.530	28.7	28.2	79	0.339	1.83
	H, p=0.3	0.158	0.526	28.0	28.4	116	0.450	2.44
	T	0.006	—	25.5	28.2	6	0.153	0.83
density, λ=0.72	F	0.174	—	18.3	17.9	56	0.243	—
	H, p=0.1	0.173	0.672	18.1	18.1	93	0.424	1.74
	H, p=0.3	0.174	0.669	18.2	18.1	174	0.852	3.51
	T	0.007	—	13.3	14.8	6	0.204	0.84
π_w^{tree}								
uniform, λ=0.10	F	0.741	—	41.05	40.93	10 035	60.6	—
	H, p=0.1	0.742	0.419	40.98	40.94	10 677	56.4	0.93
	H, p=0.3	0.744	0.422	41.05	40.90	8 010	34.2	0.56
perimeter, λ=0.34	F	0.733	—	13.04	13.04	10 257	62.3	—
	H, p=0.1	0.732	0.714	13.04	13.03	9 459	54.2	0.87
	H, p=0.3	0.735	0.715	13.06	13.04	7 686	40.2	0.64
density, λ=0.72	F	0.662	—	7.30	7.31	9 295	53.7	—
	H, p=0.1	0.662	0.831	7.34	7.27	8 334	46.0	0.86
	H, p=0.3	0.660	0.832	7.28	7.29	5 927	34.7	0.65

Table 8: Iowa short runs ($k = 4$, $\varepsilon = 0.05$, 2×10^5 steps, seed 42, $\tau = 0.1$). The two $\bar{w}$ columns are the mean weighted cut over the halves of the retained trace; "ratio" is effective samples per second relative to the forest move at the same target, timed in the same session. The partition-level tree move T is included under π_w for the baseline-against-baseline comparison; it has no tree-weighted implementation.

Long runs, the verdict. Table 9 gives 2×10^6 steps under π_w on two seeds, and 5×10^5 steps under π_w^{tree}, where the forest move's autocorrelation time is a hundred times shorter. Under π_w the pre-registered criterion passes in all four cells: ratios 1.65 and 2.29 under the perimeter target, 1.95 and 2.03 under density, with the two chains' means within 1.5 standard errors in every cell and quarter means on the same level. The forest move's effective sample sizes are 247 to 585, which puts roughly 25 percent on each ratio, so the honest statement is "1.6 to 2.3, about two". At $p_{\text{loc}} = 0.1$ the two chains cost the same

per step on this implementation, so the per-step ratios equal the per-second ones. At $p_{\text{loc}} = 0.3$ the ratios are 2.3 to 4.4 and the per-step cost ratio is 0.9 to 1.1; one level check there is borderline ($z = 2.0$, density, seed 43). Under π_w^{tree} the local move is a passenger with ratios 0.81 to 0.90 at effective sample sizes of 20 000 to 27 000, so those ratios are tight; per step the hybrid is slightly worse than the forest move there, so a local step contributes less decorrelation than the forest step it displaces before any cost is counted.

Target	Seed	ESS		ESS/s		ratio	z	quarter means	
		F	H	F	H			F	H
π_w, 2×10^6 steps, H at $p_{\text{loc}} = 0.1$									
perimeter	42	421	696	0.123	0.202	**1.65**	+0.2	28.2 27.8 28.0 28.4	28.2 28.0 28.3 28.1
perimeter	43	247	572	0.072	0.165	**2.29**	+0.2	28.6 28.2 28.7 27.9	28.3 28.6 28.5 28.2
density	42	486	925	0.177	0.345	**1.95**	−0.3	18.0 18.0 18.3 17.9	17.9 18.0 18.2 17.9
density	43	585	1 167	0.214	0.434	**2.03**	+1.5	17.9 17.8 17.7 18.2	18.0 18.0 18.0 18.1
π_w, 2×10^6 steps, H at $p_{\text{loc}} = 0.3$									
perimeter	42	421	1 116	0.123	0.324	2.65	−0.0		28.2 28.1 28.1 28.1
perimeter	43	247	1 101	0.072	0.318	4.43	−0.9		28.2 28.3 28.2 28.1
density	42	486	1 395	0.177	0.559	3.16	+0.8		18.1 18.2 18.1 18.1
density	43	585	1 244	0.214	0.497	2.33	+2.0		18.0 18.1 18.0 18.2
π_w^{tree}, 5×10^5 steps, H at $p_{\text{loc}} = 0.1$									
perimeter	42	27 019	24 558	43.9	37.4	0.85	−0.6	13.05 13.05 13.05 13.02	13.03 13.04 13.02 13.05
perimeter	43	27 113	23 604	44.1	35.8	0.81	+0.3	13.01 13.02 13.05 13.07	13.03 13.04 13.02 13.07
density	42	21 642	19 615	32.3	29.1	0.90	−1.1	7.30 7.30 7.29 7.32	7.31 7.28 7.27 7.29
density	43	22 750	19 647	33.9	29.2	0.86	−1.0	7.30 7.30 7.27 7.30	7.27 7.29 7.28 7.29

Table 9: The Iowa kill test, long runs ($k = 4$, $\varepsilon = 0.05$, $\tau = 0.1$, 10% burn-in). z is the difference of the two chains' post-burn means in standard errors computed from their own effective sample sizes. Acceptance rates: forest 0.16 (perimeter) and 0.18 (density) under π_w, 0.73 and 0.66 under π_w^{tree}; local 0.53 and 0.67 under π_w, 0.72 and 0.84 under π_w^{tree}. All ten π_w chains ran concurrently on one machine; cost-corrected ratios (effective samples per step times the single-process cost ratio of Table 8) are 1.63, 2.29, 1.99 and 2.08 at $p_{\text{loc}} = 0.1$.

Reading. Against the exact global move that practice runs, the diffusion move roughly doubles effective samples per second for geography-weighted compactness targets on Iowa and adds nothing on the tree-weighted tempered measure at the same temperature. §4.7.4 says why: at λ equivalent to 0.1 the tree-weighted target is close to the untempered tree measure the forest move proposes from, so there is nothing for a local move to add, while π_w is far from it through the tree-count term. The next subsection shows that the second case is the general one.

5.5 The scaling experiment

We run the forest move alone and the hybrid at $p_{\text{loc}} = 0.3$ on $N \times N$ grids with $k = 2$, so the merged pair is the whole graph and there is no pair-selection noise, unit populations, tolerance 0.10 (fixed in relative terms, so single flips are feasible at every N), uniform kernel, for $N \in \{6, 8, 10, 12, 16, 20\}$ and four targets: π_w at $\lambda = 1$ and 2 and π_w^{tree} at $\lambda = 0.1$ and 1. Each chain runs 5×10^4 steps, doubled for $N \geq 16$, from a vertical-split start. The forest move's acceptance and the two terms of its Hastings ratio are read from the hybrid's trace, where every forest proposal is made from a target-distributed state even at sizes where the forest move alone does not equilibrate within the budget.

Three findings. First, the exact tree move freezes as the pair grows, and the two tempered targets freeze it at the same rate. Under π_w at $\lambda = 2$ and under π_w^{tree} at $\lambda = 1$ the forest move's acceptance falls on a straight line in log scale, about $e^{-0.24N}$, from 0.30 and 0.22 at $N = 6$ to 0.009 and 0.007 at $N = 20$, while the per-proposal mismatch grows from about 2 to 19 in proportion to the mean cut length (7 to

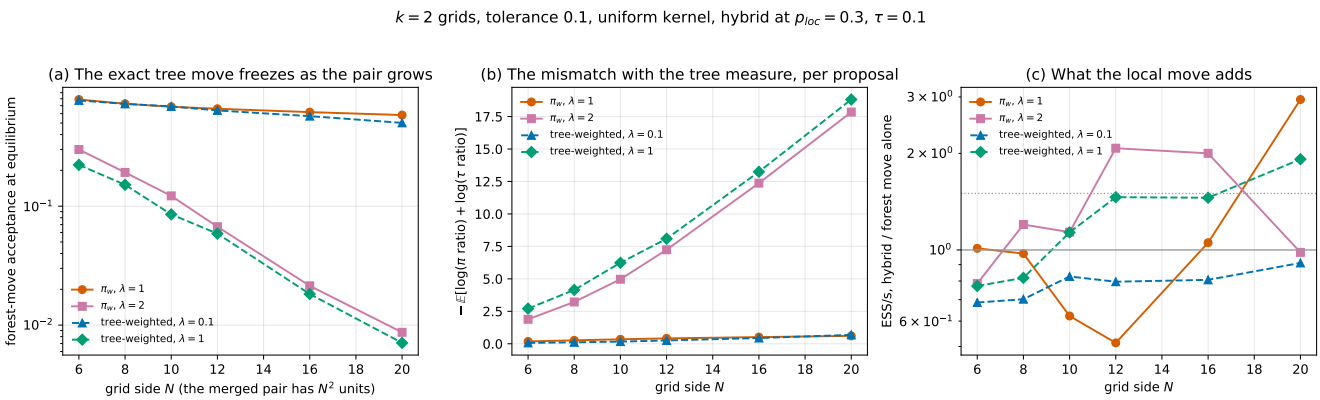


Figure 4: The exact tree move against grid side, $k = 2$, tolerance 0.1, uniform kernel, hybrid at $p_{\text{loc}} = 0.3$ and $\tau = 0.1$. (a) Forest-move acceptance at equilibrium. (b) Minus the mean of the target and tree-count log terms of (13) per feasible proposal, the per-proposal mismatch with the untempered tree measure. (c) Effective samples per second of the hybrid over the forest move alone; the $N = 20$ points of the two tempered targets are not at equilibrium (see text).

	$N = 6$	8	10	12	16	20
forest-move acceptance at equilibrium						
π_w, λ=1	0.79	0.73	0.69	0.66	0.62	0.58
π_w^{tree}, λ=0.1	0.78	0.73	0.69	0.64	0.57	0.50
π_w, λ=2	0.30	0.19	0.12	0.067	0.021	0.009
π_w^{tree}, λ=1	0.22	0.15	0.085	0.059	0.018	0.007
mean cut length under the target						
π_w, λ=2	7.2	10.0	12.6	15.1	19.8	25.1
π_w^{tree}, λ=1	6.9	9.7	12.1	15.0	20.0	25.1
per-proposal mismatch, $-\mathbb{E}[\log(\textit{target ratio}) + \log(\textit{tree-count ratio})]$						
π_w, λ=2	1.9	3.2	5.0	7.2	12.4	17.9
π_w^{tree}, λ=1	2.7	4.2	6.2	8.1	13.2	18.8
π_w, λ=1	0.2	0.3	0.3	0.4	0.5	0.6
local-move acceptance, $\tau = 0.1$ / $\tau = 0.03$						
π_w, λ=2	0.39 / 0.74	0.32 / 0.69	0.25 / 0.63	0.19 / 0.59	0.12 / 0.49	0.08 / 0.42
ESS/s hybrid over forest, $\tau = 0.03$ *(per step in parentheses)*						
π_w, λ=2	0.83 (0.78)	1.03 (0.94)	1.05 (0.91)	1.86 (1.60)	1.72 (1.46)	[3.78]
π_w^{tree}, λ=1	0.79 (0.76)	0.89 (0.80)	1.20 (1.03)	1.53 (1.31)	2.44 (2.11)	[1.32]

Table 10: The scaling experiment. Acceptance, cut length and mismatch are from the hybrid's trace at $\tau = 0.1$; the local-move acceptance and the ratios are from the $\tau = 0.03$ rerun of the two tempered targets, which the move-size law calls for once the boundary grows. Bracketed ratios are at $N = 20$, where the forest move's effective sample size is 62 and 77 and the two chains' means differ by more than two standard errors, so neither chain is at equilibrium within the budget and the ratio is not a number.

25). The two curves lie on top of each other in both panels of Figure 4, although the tree-count term is present in one Hastings ratio and absent from the other. What the move pays for is the distance of the target from the untempered tree measure, not the tree-count term as such. Under the two targets near that measure the acceptance drifts from 0.8 to 0.5 over the same range and the mismatch stays below one.

Second, the local move loses acceptance with pair size too, and has a dial for it. At $\tau = 0.1$ its acceptance falls from 0.39 to 0.08 between $N = 6$ and 20, because its expected move $2\tau|\partial_w|$ grows with the boundary and multi-unit draws are more often infeasible. At $\tau = 0.03$ it is 0.74 to 0.42 over the same range. The tree move has no such dial: its proposal is the same law at every temperature and every kernel, so its acceptance is set by the mismatch, which grows with the boundary.

Third, the local move's contribution grows with pair size where the tree move pays. At $\tau = 0.03$ the hybrid's effective samples per second relative to the forest move rise from below one at $N \leq 10$ to 1.5 to 2.4 at $N = 12$ and 16 for both tempered targets, with the chains at the same level and the per-step ratios within 15 percent of the per-second ones. At $N = 20$ neither chain reaches equilibrium in 10^5 steps under a tempered target, so that size is beyond the budget of either chain and we do not quote a ratio. Under the two near-proposal targets the ratio stays below one at every size, as on Iowa under the tree-weighted target.

What this establishes. An exact tree move serves one family of targets well, the mildly tempered tree measure, which is what recombination samples anyway. For compactness targets away from it, in either direction, its acceptance decays exponentially in the pair size, and the diffusion move, whose ratio involves a one-unit boundary change and whose reach is set by a timestep rather than by the target, is the component that keeps an exact sampler moving. The Iowa result of §5.4 is one point on these curves, at a pair size of about fifty units and a mismatch of about six.

5.6 Iowa: the symmetric-q filter against the exact sampler

On Iowa there is no enumeration, but the exact chains of §4.7 estimate both targets directly, and the forest move reaches the equilibrium of π_w within about a thousand steps where the partition-level tree move needed about 10^5. Under the uniform kernel at $\lambda = 0.1$ the forest move and the hybrid put the nominal target's mean cut count at 88 to 90 (half-run means 87.8 to 90.1 in Table 8) and the tree-weighted tempered measure's at 41.0 (half-run means 41.05 and 40.93). Tempered ReCom with the symmetric-q filter, run for 500 steps from the same kind of tree-partition start (seed 42, burn-in 100, per-step balance audit passed), sits at 39.8 ± 4.3 cuts, and Vanilla ReCom at 41.4 ± 4.8; Chain E alone, climbing from the same start, averages 55 cuts over steps 100 to 500 and ends the run at 69, still rising. On real geography the chain practitioners run as "targeting π" therefore samples plans with less than half the boundary length that π itself has, and sits within one cut of the tree-weighted tempered measure, with an exact sampler on each side of the comparison. The 5×5 calibration showed the effect at the level of distributions; Iowa shows its size.

5.7 Precinct scale

[To be written. North Carolina precinct dual graph (2 692 units, $k = 4$, $\varepsilon = 0.05$), perimeter and density targets, forest move alone and hybrid at $p_{\text{loc}} = 0.1$ with τ set by the move-size rule, 2×10^5 steps, two seeds; a timing probe first. Report forest acceptance, the two Hastings terms, quarter means, ESS per step and per second, and the ratio. Expected from §5.5: forest acceptance well below 10^{-2} under the tempered targets at pairs of about 1 300 units, hybrid reaching equilibrium where the forest move alone does not.]

6 Discussion

6.1 What each chain is for

Chain E is provably exact for its target and, alone, too immobile to be a sampler; its value is as the local component of the hybrid. **The forest move** is the mobile exact component and the baseline that practice runs; on its own measure, the mildly tempered spanning-tree distribution, it is hard to improve on, and away from it its acceptance decays with the pair size. **The hybrid** is the practical exact sampler for kernel-weighted compactness targets, with a small local fraction on real geography, and a timestep set by the move-size law rather than fixed. **The partition-level tree move** is kept as the move whose closed form explains why Chain A is only approximate. **Chains A and V** are what practitioners run, and §5.1 and §5.6 measure how far their output sits from the tempered target and how close to the tree-weighted one.

Why does the fidelity term matter? Without it the local move is a fresh spectral re-split that discards the structure the chain has found; with it the move is anchored and its size is governed by Corollary 4.1. The community step-trace diagnostic (Appendix B) makes a complementary point about locality in general. Pure tree-move chains cross community boundaries at 13–15% per step, near the community-blind baseline, while optimization-guided local mixtures drop to 5–11%, which is the division of labor the hybrid formalizes.

6.2 Cost, and what a compiled implementation would change

All chains here are Python. On Iowa under π_w a forest step costs about 1.1 ms (a Wilson draw, the balanced-edge scan, two determinants and the crossing-edge sums, with the retry multiplying the first), the partition-level tree move 0.18 ms, and a local step about 1.5 ms, half of which is solver overhead on the pairs where the balance window binds. A compiled forest step would be an order of magnitude cheaper; the local step would gain less, since the solver overhead is language independent. The per-second ratios of §5.4 would therefore move against the hybrid in Julia or Rust, which is why every comparison also reports effective samples per step. At $p_{\text{loc}} = 0.1$ the per-step and per-second ratios coincide on this implementation, so the headline result does not depend on the language. Two improvements are within reach and belong to the journal version: a closed-form solve for the local move when the balance constraint is active (the unconstrained solve shifted by a multiple of the population vector, one extra linear solve and a scalar root), which removes the solver from the local step entirely, and a per-pair timestep $\tau = m/(2|\partial_w|)$ from Corollary 4.1 that holds the expected move at m units whatever the pair, the principled form of the $\tau = 0.03$ rerun in §5.5.

6.3 On the macro-structure findings

The corrected picture of Appendix B has two consequences. First, it relocates the blame for slow local mixing. If plan space were islands joined by narrow corridors, slow traversal would be a topological inevitability; since the fibers are well connected, it is a property of the dynamics, and §4.6 identifies that property for Chain E as the feasibility of its draws, not the graph. Second, worst-case hardness results [2, 15] permit bottlenecked instances and Gold [30] exhibits planar-grid configurations that trap recombination for exponentially many expected steps, so our grids may simply be too small or too symmetric to show such structure; the statistics of Table 13 give the template for checking a fiber that does.

6.4 Limitations

(L1) No theoretical mixing-time bound. Like every redistricting chain used in practice, we lack a polynomial mixing-time guarantee (the polynomial-time grid sampler of [13] is not a local chain), and Chain E demonstrates that even an exactly reversible chain can be an extremely slow mixer. The scaling result is empirical, a straight line over six grid sizes, with the independence-sampler argument of §4.7.4 as its explanation and not its proof.

(L2) Chain A's symmetric-q approximation is not a small bias. On the enumerated fiber the tempered chain's empirical mean (16.73) sits 1.8 cuts below the exact target mean (18.56), its cut distribution is 220 noise-floor standard deviations from that target (§5.1), and on Iowa the gap is a factor of two in cut length (§5.6).

(L3) Chain E mixes too slowly to use alone. Exactness comes from single-shot rounding, whose reach Corollary 4.1 bounds and whose feasibility §4.6 measures. Alone it is not a sampler; in the hybrid it is a component whose value depends on how far the target sits from the tree measure.

(L4) The Iowa ratios carry about 25 percent uncertainty and rest on two seeds. The forest move's effective sample size under π_w is in the hundreds at 2×10^6 steps. The direction, the level agreement and the control are solid; the second decimal of the ratio is not.

(L5) Scale. The largest real graph is 99 counties. §5.7 is to supply the precinct-scale run; until then the scaling experiment on grids is the evidence that the effect grows with pair size.

6.5 Future directions

1. The precinct-scale run of §5.7, and the downstream question it allows: whether the choice of kernel changes an ensemble's partisan statistics on real elections.

2. The integer form of the energy, the same objective over $x \in \{0,1\}^{|V_H|}$ solved as a convex mixed-integer program inside a tempered chain, is an anchored refinement tool rather than a sampler. In an earlier version of this work we evaluated it as an optimizer on Iowa: it reduced mean cut edges by 20 percent over tempered ReCom, and simulated annealing over ReCom proposals reached the same best plans at a fraction of the cost. We record the result so that it need not be rediscovered, and do not pursue it.

3. An informed marked-edge selection for the marked edge walk [38], with a Boltzmann draw over marked sets whose density is computable, would be the lifted-space analogue of Chain E; the same single-shot principle applies.

4. Integration with the cycle walk [24] or the balanced up-down walk [1] as the global component in place of the forest move, and multi-scale diffusion kernels in the local move.

7 Conclusion

This paper introduced a diffusion proposal for redistricting Markov chains, one Laplacian energy $\alpha\, x^\top L x + \beta \,\|x - x_0\|^2$ under global population balance, and showed that the diffusion view is more than an interpretation. The proposal is a single linear solve whenever balance is slack, its expected move size is an explicit filter norm with a small-timestep law and a hard ceiling, and its mobility is governed by feasibility rather than by the Metropolis ratio. Corrected by an exact Metropolis–Hastings step, the proposal gives Chain E, to our knowledge the first optimization-based proposal in redistricting MCMC with a detailed-balance argument, and on an 18-state fiber a million-step run's occupancy distribution is not distinguishable from the exact target. Lifted to the spanning-forest state space and mixed with the

forest-lifted recombination move Metropolized to the same target, it gives a reversible sampler for kernel-weighted compactness targets, validated on 4 006 enumerated plans, that on Iowa roughly doubles the effective samples per second of the forest move under perimeter and density targets and adds nothing on the tree-weighted tempered measure. The reason is structural and measured: an exact tree move proposes from the untempered spanning-tree measure whatever the target, its acceptance decays exponentially in the pair size under any compactness-tempered target, and the local move has a dial for the same effect where the tree move has none. On the enumerated 5×5 fiber we showed at the level of distributions that tempered ReCom does not sample its nominal target and sits close to the spanning-tree-weighted measure, and on Iowa that the gap is a factor of two in boundary length.

For the field, we hope three ideas outlast the specific chains. Kernel choice can be a first-class modeling decision inside the proposal and the target rather than a post-hoc filter, provided the induced distribution is corrected. The spanning-tree measure is the one target a tree move samples for free, and every other target costs it an acceptance that shrinks with the units in a pair, so a sampler for compactness targets at precinct scale needs a move whose reach does not depend on the target. And enumerated fibers are cheap, severe test beds; a chain's distance from its own computable target on 4 006 states, read against a noise floor, is a more informative diagnostic than any p-value on a single trace.

Code and data availability

All code, the enumerated fibers, and every artifact behind every table and figure are in the public repository `https://github.com/kirtisoglu/Redistricting_Diffusion_Model`, under `diffusion_model/recom/`, with the scripts and command lines listed in Appendix C. An archived release with a persistent identifier will be deposited on acceptance.

A Algorithm details and hyperparameters

A.1 Common setup

- Dual graph $G = (V, E)$ with vertex populations p_v and edge attributes (`shared_perim` for Iowa).
- k districts, ε-population balance per Equation (2), enforced as a two-sided bound around the global ideal on every move (Equation (4)) and audited per step. The bare GerryChain `recom` proposal is wrapped with an explicit whole-plan balance check (`recom_valid`), since its internal cut search validates only one side of each split.
- Initial partition from `gerrychain.tree.recursive_tree_part`, redrawn until all k districts satisfy the global tolerance (`balanced_tree_part`); vertical split for the $k = 2$ grids.
- Step counts: 5×5 baselines $N = 1\,000$, burn-in 100; Iowa baselines $N = 500$, burn-in 100; long-run validation (§5.1) 2×10^5 (A) and 10^5 (E); 5×5 forest-move validation (§5.3) 2×10^5; Iowa short runs 2×10^5, long runs 2×10^6 under π_w and 5×10^5 under π_w^{tree}; scaling experiment 5×10^4 per chain, 10^5 for $N \geq 16$. Burn-in 10% unless stated.
- Random seeds: base seed 42 with a distinct offset per chain (independent streams for Python `random` and `numpy`); seed 43 for the second Iowa long run.

A.2 Chain-specific parameters

Table 11 lists every locked value.

Chain	Parameter	Value
A	λ (target temperature)	0.10
E	α, β	1.0, 10.0 ($\tau = \alpha/\beta = 0.1$)
E	Rounding	single-shot Bernoulli($x^\star$), clip 10^{-12}
T	cut edge	uniform over all tree edges; fair-coin labels
F	cut edge	uniform over balanced tree edges; fair-coin labels; retry up to 20 trees
F	pair	uniform over boundary pairs
H	p_{loc}	0.1 (Iowa long runs), 0.1 and 0.3 (Iowa short runs), 0.2 (5×5), 0.3 (grids)
H	τ	0.1; 0.03 in the rerun of §5.5
Iowa λ	uniform / perimeter / density	0.100 / 0.343 / 0.718 (equal tempering at the start plan)

Table 11: Locked-in hyperparameters for the chains of Table 1.

A.3 Compute environment

- macOS (Darwin 23.6.0), Python 3.12, and a Linux container (Python 3.13) for the analytic checks and prototypes.
- Gurobi 13.0 (academic license) for the QP solves that the linear-solve path of Lemma 4.1 does not cover.
- GerryChain 0.2.22 for the ReCom proposal and the initial tree partitions; NetworkX 3.x for graph operations.
- One Python process per chain, single-threaded, at reduced priority; the ten Iowa long runs and the scaling chains ran concurrently on an 11-core machine, each forest chain at the same time as its hybrid partner so that both members of a ratio see the same load. All ratios are also reported per step.

B Partition fiber structure: communities without bottlenecks

This appendix documents a structural diagnostic of the partition fiber $\mathcal{P}_k(G;\varepsilon)$ (the set of all valid plans), viewed as a graph under its minimal valid move relation. Two findings emerge, one positive and one negative, and we report both. Positively, all three enumerated fibers carry genuine mesoscale community structure, with modularity far above degree-preserving null models, and chains differ sharply in how often they cross community boundaries. Negatively, a plausible folk intuition (that the communities are "islands" joined by narrow corridors or bottleneck edges) is refuted by method-independent statistics. Community interfaces are wide, nearly all community pairs are directly connected, and the literal bridges of the exact-balance fiber turn out to be pendant vertices. The diagnostic still motivates the local-refinement design of §4, and we believe both findings are of independent interest for analyzing redistricting MCMC.

Three test beds. We construct three fully enumerated partition fibers spanning grid size, district count, and balance tolerance (Table 12).

For each fiber, all valid partitions are enumerated and the single-flip adjacency graph is built explicitly (enumeration runtimes on a 2024 MacBook range from ~ 5 s for the 4×4, $\varepsilon = 0.25$ fiber to ~ 50 min for the 5×5, $\varepsilon = 0.20$ fiber; scripts accompany the replication materials).

Community structure, validated against null models. Each fiber graph carries substantial mesoscale community structure under modularity maximization. We report the method per fiber (greedy

Fiber	Nodes	Edges	k	ε	Sizes	Adjacency
5×5, $\varepsilon = 0$	4 006	13 416	5	0	{5}	boundary swap
5×5, $\varepsilon = 0.20$	193 128	969 760	5	0.20	{4, 5, 6}	single flip
4×4, $\varepsilon = 0.25$	1 953	7 024	4	0.25	{3, 4, 5}	single flip

Table 12: The three test-bed fibers. Adjacency is the *minimal valid move* of each fiber. At $\varepsilon = 0$ no single-vertex reassignment preserves exact balance (it produces districts of sizes 4 and 6), so adjacency is the boundary *swap*, which exchanges one boundary vertex of district a with one of district b, both districts remaining connected. At $\varepsilon > 0$ adjacency is the single *flip*, which reassigns one boundary vertex, both affected districts remaining connected and within size bounds. Cross-fiber comparisons carry one caveat. The $\varepsilon = 0$ fiber lives under a different (two-vertex) move relation than the $\varepsilon > 0$ fibers.

modularity, Clauset–Newman–Moore, for the $\varepsilon = 0$ and 4×4 fibers; Louvain, resolution 1.0, seed 42, for the large $\varepsilon = 0.20$ fiber), validate against degree-preserving rewiring nulls, and treat the specific community counts as method-dependent descriptive statistics.

	Community structure				Anti-bottleneck statistics		
Fiber	Comm.	Q	Null Q	Method	Pairs conn.	Inter-comm. edges	Width (med/max)
5×5, $\varepsilon = 0$	10	0.65	0.36	CNM	15/15	18.3%	142 / 383
5×5, $\varepsilon = 0.20$	26	0.69	n/a	Louvain	320/325	26.8%	667 / 3 006
4×4, $\varepsilon = 0.25$	7	0.58	0.34	CNM	21/21	23.3%	71 / 186

Table 13: Left: community decompositions with degree-preserving null baselines (mean Q of CNM on double-edge-swap rewirings of the same graph; the 193k-node fiber's null is omitted for cost). Real Q exceeds null Q by 0.24–0.29, so the community structure is genuine, not a degree-sequence artifact. Counts vary with the detection method and should be read qualitatively. Right: the statistics that refute the corridor picture. Nearly every pair of communities shares a direct interface ("pairs conn." counts the large-community pairs at $\varepsilon = 0$), interfaces are tens to thousands of edges wide, and a fifth to a quarter of all fiber edges run between communities. The exact-balance fiber's 16 literal single-edge bridges each split off a *single vertex* (pendant leaves), not an island, and the six large $\varepsilon = 0$ communities have conductance 0.15–0.22, an order of magnitude above bottleneck territory.

No narrow corridors: a corrected intuition. It is tempting to picture these communities as islands joined by a few narrow corridors, with bottleneck edges controlling passage. The method-independent statistics on the right of Table 13 refute that picture on all three fibers.

Concretely, on the $\varepsilon = 0.20$ fiber, interfaces thinner than 50 edges do exist (there are 16 of them), but they are simply the tail of a width distribution whose median is 667 and whose top interfaces carry 3 000 edges; a chain moving through this graph crosses between communities through the wide interfaces, and the thin tail is irrelevant to mixing. The correct summary of fiber macro-structure is therefore *communities without bottlenecks*. Valid-plan space is regionally organized but well connected, and slow traversal by local chains (below) is a property of the *dynamics*, not of corridor topology. This negative result matters for chain design. It says local chains stall not because the state space is nearly disconnected but because their moves are small and, in Chain E's case, because distant proposals are rarely feasible (§4.6); global recombination moves, by contrast, relocate across communities in a single step.

Chain step-trace diagnostic. The community decomposition supports a direct diagnostic for any redistricting chain. We label each visited partition by its greedy-modularity community on the $\varepsilon = 0$ fiber

and record the *jump rate*, the fraction of steps that move to a different community (Table 14).

Chain	Communities visited	Jump rate	Jumps in 1 000 steps
V: Vanilla ReCom	8/10	15.3%	153
A: Tempered ReCom	7/10	12.7%	127
B: continuous QP mix (50/50)	6/10	4.6%	46
C: integer QP refinement	6/10	10.5%	105
D: ReCom + spectral mix	7/10	9.3%	93
E: QP-MH exact	2/10	0.3%	3

Table 14: Inter-community jump rates on the 5×5, $\varepsilon = 0$ fiber (1 000 steps per chain, seed 42, per-chain RNG streams). A community-blind baseline crosses at roughly the inter-community edge share (18.3%, Table 13); ReCom variants (V, A) sit near it at $\sim$ 13–15% per step, the local kernels (B, C, D) are stickier at $\sim$ 5–11%, and the strictly local QP-MH chain (E) almost never crosses, a property of its small accepted moves rather than of any bottleneck in the fiber. Chains B, C and D are the deterministic-rounding mixtures of an earlier version of this work (§4.4, §6.5), included because the diagnostic was run on all six.

Reading of the diagnostic. ReCom relocates across communities in single steps; optimization-guided local kernels work within them; and the exact-but-strictly-local Chain E shows the failure mode of small accepted moves. The regimes are complementary, and they suggest an explicit design principle. A good redistricting sampler should combine global moves that relocate across regions of plan space with local moves that refine within them; the forest move does the former, the anchored local kernel the latter. This is the design of the hybrid of §4.7, each step a random mixture of a tree move and a diffusion move. We note the diagnostic's limits (one run per chain on one fiber, and community labels from a specific detection method) and read it as motivating evidence, not proof.

C Reproducibility checklist

- Code: `diffusion_model/recom/` in the project repository.
- Modules: `problems.py` (graph setup, balanced initial plans), `chains.py` (chain runners, balance-validated ReCom wrapper, per-step audit), `qp_model.py` (QP and QP-MH step factories, global-ideal balance windows), `hybrid.py` (the partition-level tree move, Chain E with the linear-solve path, and their analytic check), `forest_recom.py` (the forest move, the lifted local move, the two targets, and the analytic check on the lifted 2×3 fiber), `kill_stats.py` (ESS, noise floors, batch-means Hotelling with bin merging), `diagnostics.py`.
- Orchestrators: `iowa_target_matched.py` (the baseline chains V, A and E), `long_run_5x5.py` and `matrix_tree_targets.py` (§5.1), `calibration_distance.py` and `fig_calibration.py` (§5.1), `db_check_2x3.py` (Remark 4.1), `long_run_2x3.py` with `gof_2x3.py` (§5.2; arguments `1000000 42`), `spectral_move_size.py` and `fig_spectral.py` (§4.3), `acceptance_sweep.py` (Table 2), `mobility_law.py` (Table 3), `forest_validate_5x5.py` (Table 7; arguments `200000 42 0.2 0.1 pi` and `... tree`), `forest_iowa.py` (Tables 8 and 9; `short 200000 42 0.1 pi|tree`, `long forest|hybrid 2000000 42|43 pi [0.3]`, `long forest|hybrid 500000 42|43 tree`), `forest_verdict.py` (the verdict table), `forest_scaling.py` with `forest_scaling_parallel.py` (Table 10, Figure 4; arguments `50000 42 6,8,10,12,16,20 pi:1,pi:2,tree:0.1,tree:1 0.3 --fig`, and the same with `TAU = 0.03` for the rerun), `hybrid_validate_5x5.py`, `hybrid_iowa.py`

and `hybrid_iowa_long.py` (the partition-level tree move's rows), `ten_seed_stats.py` (between-seed statistics).

- Outputs, all under `iowa_hybrid_test_out/`, each stamped with its configuration: `target_matched_*.npz`, `long_run_5x5_seed42.npz`, `matrix_tree_targets.npz`, `calibration_distance_seed42.npz`, `long_run_2x3_n*_seed42.npz`, `spectral_move_size.npz`, `acceptance_sweep_5x5.npz`, `mobility_law_5x5.npz`, `hybrid_*.npz`, `forest_validate_5x5_*.npz`, `forest_iowa_*.npz`, `forest_verdict.txt`, `forest_scaling_seed42*.npz`, with the console logs beside them.

- Fibers: `fiber_5x5/` contains the 5×5 and 4×4 enumeration scripts and precomputed adjacency files.

- Reproducibility across sessions. The 5×5 runs and the Iowa runs under the uniform and density kernels reproduce bit for bit from their seeds. The Iowa runs under the perimeter kernel do not: a repeat of one $\tau = 0.1$ session matched the stored trajectories for 21 217 and 185 390 steps and then diverged, which we attribute to floating-point sensitivity of the QP solver fallback under that kernel's wide range of edge weights. The stored artifacts are the realizations reported, and every perimeter-target conclusion rests on long runs and two seeds.

- Precinct scale: [data file, commands and artifacts of §5.7.]

References

[1] Hugo A Akitaya, Sarah Cannon, Gregory Herschlag, Gabe Schoenbach, Kristopher Tapp, and Jamie Tucker-Foltz. The Balanced Up-Down Walk. *arXiv preprint arXiv:2602.11993*, 2026.

[2] Hugo A. Akitaya, Matias Korman, Oliver Korten, Diane L. Souvaine, and Csaba D. Tóth. Reconfiguration of connected graph partitions via recombination. *Theoretical Computer Science*, 923:13–26, 2022.

[3] Eric A. Autry, Daniel Carter, Gregory J. Herschlag, Zach Hunter, and Jonathan C. Mattingly. Metropolized multiscale forest recombination for redistricting. *Multiscale Modeling & Simulation*, 19(4):1885–1914, 2021.

[4] Eric A. Autry, Daniel Carter, Gregory J. Herschlag, Zach Hunter, and Jonathan C. Mattingly. Metropolized forest recombination for Monte Carlo sampling of graph partitions. *SIAM Journal on Applied Mathematics*, 83(4):1366–1391, 2023.

[5] Johnathan M. Bardsley, Antti Solonen, Heikki Haario, and Marko Laine. Randomize-then-optimize: A method for sampling from posterior distributions in nonlinear inverse problems. *SIAM Journal on Scientific Computing*, 36(4):A1895–A1910, 2014.

[6] Amariah Becker and Justin Solomon. Redistricting algorithms. In Moon Duchin and Olivia Walch, editors, *Political Geometry: Rethinking Redistricting in the US with Math, Law, and Everything In Between*, pages 303–340. Birkhäuser, 2022.

[7] Andrea L Bertozzi and Arjuna Flenner. Diffuse interface models on graphs for classification of high dimensional data. *Multiscale Modeling & Simulation*, 10(3):1090–1118, 2012.

[8] Daniel Brous and David B. Shmoys. Optimizing districting plans to maximize majority-minority districts via IPs and local search. In *2025 Proceedings of the Conference on Applied and Computational Discrete Algorithms (ACDA)*, pages 264–275, 2025. arXiv:2508.07446.

[9] Sarah Cannon. Irreducibility of recombination Markov chains in the triangular lattice. *Discrete Applied Mathematics*, 347:75–130, 2024.

[10] Sarah Cannon, Daryl DeFord, and Moon Duchin. Repetition effects in a sequential Monte Carlo sampler. *arXiv preprint arXiv:2409.19017*, 2024.

[11] Sarah Cannon, Moon Duchin, Dana Randall, and Parker Rule. Spanning trees and redistricting: New methods for sampling and validation. *SIAM Review*, 68(2):349–381, 2026. arXiv:2210.01401 (2022).

[12] Sarah Cannon, Ari Goldbloom-Helzner, Varun Gupta, JN Matthews, and Bhushan Suwal. Voting Rights, Markov chains, and optimization by short bursts. *Methodology and Computing in Applied Probability*, 25(1):36, 2023.

[13] Sarah Cannon, Wesley Pegden, and Jamie Tucker-Foltz. Sampling balanced forests of grids in polynomial time. In *Proceedings of the 56th Annual ACM Symposium on Theory of Computing (STOC '24)*, pages 1676–1687, 2024.

[14] Tony F Chan and Luminita A Vese. Active contours without edges. *IEEE Transactions on image processing*, 10(2):266–277, 2001.

[15] Moses Charikar, Paul Liu, Tianyu Liu, and Thuy-Duong Vuong. On the complexity of sampling redistricting plans. *arXiv preprint arXiv:2206.04883*, 2022.

[16] Gabriel Chuang, Gregory Herschlag, and Jonathan C. Mattingly. Multiscale parallel tempering for fast sampling on redistricting plans. *Multiscale Modeling & Simulation*, 23(4):1515–1550, 2025.

[17] Fan RK Chung. *Spectral graph theory*, volume 92. American Mathematical Soc., 1997.

[18] Jeanne N. Clelland, Nicholas Bossenbroek, Thomas Heckmaster, Adam Nelson, Peter Rock, and Jade VanAusdall. Compactness statistics for spanning tree recombination. *arXiv preprint arXiv:2103.02699*, 2021.

[19] Jeanne N. Clelland, Haley Colgate, Daryl DeFord, Beth Malmskog, and Flavia Sancier-Barbosa. Colorado in context: Congressional redistricting and competing fairness criteria in Colorado. *Journal of Computational Social Science*, 5(1):189–226, 2022.

[20] Jeanne N. Clelland and Kristopher Tapp. Bonsai: A class of effective methods for independent sampling of graph partitions. *arXiv preprint arXiv:2603.18347*, 2026.

[21] Ewan Davies, Ryan Job, Maxine Kampbell, Hannah Kim, and Hyojin Seo. Spectral partitioning of graphs into compact, connected regions. *arXiv preprint arXiv:2506.13982*, 2025.

[22] Daryl DeFord, Moon Duchin, and Justin Solomon. Recombination: A family of Markov chains for redistricting. *Harvard Data Science Review*, 3(1), 2021.

[23] Daryl DeFord, Hugo Lavenant, Zachary Schutzman, and Justin Solomon. Total variation isoperimetric profiles. *SIAM Journal on Applied Algebra and Geometry*, 3(4):585–613, 2019.

[24] Daryl R DeFord, Gregory Herschlag, and Jonathan C Mattingly. A cycle walk for sampling measures on spanning forests for redistricting. *arXiv preprint arXiv:2509.08629*, 2025.

[25] Kiera W. Dobbs, Douglas M. King, Ian G. Ludden, and Sheldon H. Jacobson. Facilitating compromise in redistricting with transfer distance midpoints. *INFORMS Journal on Optimization*, 6(3–4):214–239, 2024.

[26] Moon Duchin and Bridget Eileen Tenner. Discrete geometry for electoral geography. *arXiv preprint arXiv:1808.05860*, 2018.

[27] Benjamin Fifield, Michael Higgins, Kosuke Imai, and Alexander Tarr. Automated redistricting simulation using Markov chain Monte Carlo. *Journal of Computational and Graphical Statistics*, 29(4):715–728, 2020.

[28] Benjamin Fifield, Kosuke Imai, Jun Kawahara, and Christopher T Kenny. The essential role of empirical validation in legislative redistricting simulation. *Statistics and Public Policy*, 7(1):52–68, 2020.

[29] Matteo Fischetti and Andrea Lodi. Local branching. *Mathematical Programming*, 98(1–3):23–47, 2003.

[30] Micah Gold. Adversarial configurations for the ReCom transition function. *arXiv preprint arXiv:2606.01333*, 2026.

[31] Wes Gurnee and David B. Shmoys. Fairmandering: A column generation heuristic for fairness-optimized political districting. In *SIAM Conference on Applied and Computational Discrete Algorithms (ACDA21)*, pages 88–99, 2021.

[32] Gregory Herschlag, Han Sung Kang, Justin Luo, Christy Vaughn Graves, Sachet Bangia, Robert Ravier, and Jonathan C. Mattingly. Quantifying gerrymandering in North Carolina. *Statistics and Public Policy*, 7(1):30–38, 2020.

[33] Matt Jacobs, Ekaterina Merkurjev, and Selim Esedoğlu. Auction dynamics: A volume constrained MBO scheme. *Journal of Computational Physics*, 354:288–310, 2018.

[34] Matt Jacobs and Olivia Walch. A partial differential equations approach to defeating partisan gerrymandering. *arXiv preprint arXiv:1806.07725*, 2018.

[35] David R Karger. Minimum cuts in near-linear time. *Journal of the ACM (JACM)*, 47(1):46–76, 2000.

[36] Cory McCartan and Kosuke Imai. Sequential Monte Carlo for sampling balanced and compact redistricting plans. *The Annals of Applied Statistics*, 17(4):3300–3323, 2023.

[37] Cory McCartan, Christopher T. Kenny, Tyler Simko, George Garcia, Kevin Wang, Melissa Wu, Shiro Kuriwaki, and Kosuke Imai. Simulated redistricting plans for the analysis and evaluation of redistricting in the United States. *Scientific Data*, 9:689, 2022.

[38] Atticus McWhorter and Daryl DeFord. The marked edge walk: A novel MCMC algorithm for sampling of graph partitions. *arXiv preprint arXiv:2510.17714*, 2025.

[39] Ekaterina Merkurjev, Tijana Kostic, and Andrea L Bertozzi. An MBO scheme on graphs for classification and image processing. *SIAM Journal on Imaging Sciences*, 6(4):1903–1930, 2013.

[40] Metric Geometry and Gerrymandering Group. GerryChain: A versatile tool for analyzing redistricting (v0.3.1, region-aware ReCom via `region_surcharge`). https://gerrychain.readthedocs.io, 2024. Software.

[41] Elle Najt, Daryl DeFord, and Justin Solomon. Empirical sampling of connected graph partitions for redistricting. *Physical Review E*, 104(6):064130, 2021.

[42] Philip O'Sullivan, Kosuke Imai, and Cory McCartan. Generalized sequential Monte Carlo sampling for redistricting simulation. *arXiv preprint arXiv:2603.22188*, 2026.

[43] Yusuf Yiğit Pilavcı, Pierre-Olivier Amblard, Simon Barthelmé, and Nicolas Tremblay. Graph tikhonov regularization and interpolation via random spanning forests. *IEEE transactions on Signal and Information Processing over Networks*, 7:359–374, 2021.

[44] Ariel D. Procaccia and Jamie Tucker-Foltz. Compact redistricting plans have many spanning trees. In *Proceedings of the 2022 ACM-SIAM Symposium on Discrete Algorithms (SODA)*, pages 3754–3771, 2022. arXiv:2109.13394.

[45] Jianbo Shi and Jitendra Malik. Normalized cuts and image segmentation. *IEEE Transactions on pattern analysis and machine intelligence*, 22(8):888–905, 2000.

[46] David I Shuman, Sunil K Narang, Pascal Frossard, Antonio Ortega, and Pierre Vandergheynst. The emerging field of signal processing on graphs: Extending high-dimensional data analysis to networks and other irregular domains. *IEEE signal processing magazine*, 30(3):83–98, 2013.

[47] Mechthild Stoer and Frank Wagner. A simple min-cut algorithm. *Journal of the ACM (JACM)*, 44(4):585–591, 1997.

[48] Rahul Swamy, Douglas M. King, and Sheldon H. Jacobson. Multiobjective optimization for politically fair districting: A scalable multilevel approach. *Operations Research*, 71(2):536–562, 2023.

[49] Kristopher Tapp. Revisiting compactness for district plans. *arXiv preprint arXiv:2501.02325*, 2025.

[50] Kristopher Tapp, Todd Proebsting, and Alec Ramsay. Parameter effects in ReCom ensembles. *arXiv preprint arXiv:2505.21326*, 2025.

[51] Luke Tierney. A note on Metropolis–Hastings kernels for general state spaces. *The Annals of Applied Probability*, 8(1):1–9, 1998.

[52] Hamidreza Validi and Austin Buchanan. Political districting to minimize cut edges. *Mathematical Programming Computation*, 14:623–672, 2022.

[53] Hamidreza Validi, Austin Buchanan, and Eugene Lykhovyd. Imposing contiguity constraints in political districting models. *Operations Research*, 70(2):867–892, 2022.

[54] Dorothea Wagner and Frank Wagner. Between min cut and graph bisection. In *International Symposium on Mathematical Foundations of Computer Science*, pages 744–750. Springer, 1993.

[55] David Bruce Wilson. Generating random spanning trees more quickly than the cover time. In *Proceedings of the twenty-eighth annual ACM symposium on Theory of computing*, pages 296–303, 1996.

[56] Giacomo Zanella. Informed proposals for local MCMC in discrete spaces. *Journal of the American Statistical Association*, 115(530):852–865, 2020.